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Review

Multi-sensor fusion on autonomous driving perception: A comprehensive survey

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ABSTRACT

In recent years, the rapid evolution of artificial intelligence has propelled autonomous driving (AD) to the forefront of technological applications in the transportation sector, drawing significant research attention. Perception tasks have received extensive attention due to the strides in sensing technologies like cameras and Light Detection and Ranging (LiDAR). These advancements have significantly improved the accuracy and robustness of perception systems. However, the inherent limitations of single-modal sensors have necessitated a shift towards multi-modal sensor fusion. This approach leverages the unique advantages of diverse sensor systems integrated into autonomous vehicles (AVs), significantly improving perception accuracy. This paper reviews recent developments in multi-sensor fusion for AD, emphasizing its role in overcoming the limitations of single-modal sensors. It examines major open-source datasets, evaluates single-sensor perception performance, and categorizes state-of-the-art fusion algorithms from both methodological and task-oriented perspectives. The review identifies current research gaps and proposes future directions to enhance AD perception systems.

1. Introduction

In recent years, the adoption of deep learning in computer vision has significantly accelerated the development of autonomous driving (AD) technologies, promoted the advancement of intelligent transportation systems and improved driving automation capabilities. Driving automation is categorized into levels ranging from 0 to 5 (Yurtsever et al., 2020), with higher levels aiming to achieve enhanced autonomy, safety, and efficiency. As AD technology continues to evolve, researchers are increasingly focusing on enhancing the generalization capability and robustness of AD systems under dynamic, heterogeneous and uncertain real-world driving environments (Nigam and Srivastava, 2023; Xu and Zheng, 2024). For example, in mixed traffic flows involving various vehicles, pedestrians, and unpredictable road conditions (Wang et al., 2025; Zhu and Zhang, 2018), autonomous vehicles (AVs) must accurately identify

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both dynamic and static obstacles and make adaptive driving decisions in real time. Moreover, since AD algorithms govern the perception, planning, and control processes of autonomous agents, how to achieve large-scale deployment remains challenging due to reliability, computational efficiency, and safety constraints. Considering factors such as real-time operation and energy efficiency, recent research has also focused on ensuring compatibility between system software and vehicle hardware. With the planning and construction of intelligent transportation systems, technologies such as intelligent traffic signal systems and Vehicle-to-Everything (V2X) communication play a vital role in supporting the safe and efficient operation of AD agents (Liu et al., 2023). Despite the rapid progress of AD technologies and the ambition to achieve human-level or even superhuman driving performance, perception performance remains one of the major obstacles limiting the practical deployment of AD systems.

Perception constitutes the foundation of AD systems, directly supporting subsequent planning and decision-making processes. Within the scope of microscopic traffic analysis, perception tasks primarily focus on understanding the environment around the vehicle. This involves object detection (Arnold et al., 2019; Chen et al., 2017; Feng et al., 2022; Ma et al., 2024; Qian et al., 2022; Rajaram et al., 2016) and semantic segmentation (Li et al., 2023; Muhammad et al., 2022; Siam et al., 2018; Wang et al., 2022; Xiao et al., 2023; Zhou et al., 2020). Object detection localizes and classifies static and dynamic obstacles such as pedestrians and vehicles, typically producing bounding boxes or 3D extents. Object tracking complements detection by maintaining object identities over time and estimating motion states such as velocity and trajectories. Meanwhile, semantic segmentation extracts semantic scene information from scene components such as drivable areas, lane markings, traffic signs, and traffic lights. Such insights are critical for decision-making and control by autonomous agents, requiring the deployment of various onboard sensors. AVs are typically equipped with an array of sensors, including cameras, Light Detection and Ranging (LiDAR), and millimeter-wave radar (MMWR), to facilitate various perception tasks (Bilik et al., 2019). RGB cameras generate two-dimensional (2D) images that are critical for extracting semantic information from the road environment, extensively utilized in semantic segmentation and classification tasks. However, the lack of depth perception in 2D images necessitates the use of LiDAR point clouds (Liang et al., 2022), which provide depth data essential for three-dimensional (3D) object detection, lane detection, and vehicle tracking. Because LiDAR performance can degrade under adverse weather conditions and its deployment cost remains high, MMWR is increasingly employed. MMWR offers longer detection ranges and improved resistance to environmental interference, presenting a more viable option for diverse operational scenarios (Zhao et al., 2024).

Although different sensing modalities provide complementary perception capabilities, relying solely on a single sensor remains insufficient for robust autonomous driving perception under complex and dynamic real-world environments. Single-modality perception systems often struggle to simultaneously achieve reliable semantic understanding, accurate spatial perception, environmental robustness, and stable performance under adverse conditions. Consequently, multi-sensor fusion has emerged as a key research direction for integrating complementary sensing information and improving the robustness, reliability, and safety of autonomous driving perception systems.

Regarding the performance of specific sensors in different applications, Table 1 shows that the sensing ability of single-mode sensors is limited under complex environmental conditions. Integrating multiple sensors to enhance AD perception presents significant opportunities and challenges (Huang et al., 2022). Key issues include reconciling differing spatial coordinate systems and data types among sensors, which complicates the process beyond straightforward sensor aggregation, creating unique challenges for multi-sensor fusion. Recent advancements in this area have led to substantial achievements in multi-modal feature integration, model robustness, and performance optimization. However, comprehensive reviews that effectively categorize and integrate methods are lacking, with only a few review articles exploring multi-modal open-source datasets and sensor-level contributions. This limitation hinders comprehensive comparison and analysis of established methods.

Given the importance of multi-sensor fusion in AD perception, and the lack of systematic summaries and categorization in current research, this paper provides a comprehensive survey of multi-sensor fusion methods from the perspectives of datasets, single-sensor perception limitations, and representative fusion frameworks. In particular, this review proposes a unified categorization framework from both optimization-oriented and task-oriented perspectives, aiming to provide a clearer understanding of the development and application of multi-sensor fusion technologies in AD perception systems. The main contributions of this review are summarized as follows: Firstly, as the foundation of multi-sensor fusion technologies for autonomous driving perception, this paper systematically reviews representative open-source datasets, with particular emphasis on their sensor configurations, scenario diversity, environmental conditions, and supported perception tasks. The review highlights the importance of multi-modal data collection for improving the robustness and generalization capability of AD perception systems under complex real-world environments. Secondly, this review comprehensively summarizes existing perception methods based on cameras, LiDAR, and MMWR sensors, analyzing their respective strengths, limitations, and application characteristics in autonomous driving tasks. In particular, the limitations of single-modality perception systems, such as insufficient environmental adaptability, sensitivity to adverse weather conditions, and restricted perception capability under complex scenarios, are further discussed to motivate the necessity of multi-sensor fusion.

Thirdly, this paper systematically reviews and categorizes state-of-the-art multi-sensor fusion methods from both optimization-oriented and task-oriented perspectives. Building on conventional early and late fusion classification (Wang et al., 2023), it proposes a new classification based on optimization techniques, such as enhancements within the convolutional neural network (CNN) framework, attention mechanism modules, and Transformer encoder-decoder modules. The review analyzes various fusion methods through their applications in different subdivided perception sub-tasks, summarizing the strengths and limitations of current fusion algorithms from a task-oriented viewpoint. Finally, this review summarizes the major challenges and limitations of current multi-sensor fusion technologies, including data alignment, heterogeneous feature representation, computational efficiency, and robustness under adverse environments. Potential future research directions are further discussed to provide insights for the development of more efficient, reliable, and robust autonomous driving perception systems.

Table 1
A Performance Comparison of the Sensors and the Degree of Performance Improvement of Multi-sensor Fusion.

Sensor Performance	MMWR	LiDAR	Camera	MMWR + Camera	LiDAR + Camera	MMWR + LiDAR	MMWR + LiDAR + Camera
Sensing Range				--	--	--	--
Robustness to Illumination Intensity				↑↑	↑	↑	↑↑
Robustness to Adverse Weather				↑↑	--	↑	↑↑
Object Classification Performance				↑↑	↑↑	--	↑↑
Object Detection Performance				↑↑	↑↑	↑	↑↑
Velocity Measurement				↑	↑	/	↑↑
Semantic Information Representation				↑↑	↑↑	/	↑↑
Sensor Cost				↓	↓↓	↓↓↓	↓↓↓↓

Note: Green, yellow, and red represent performance levels ranked from high to low, respectively. The symbols "↑" and "↓" denote the improvement and degradation of sensor performance, respectively, relative to the optimal single-sensor performance prior to fusion, based on the overall performance trends reported in the corresponding studies. "-" indicates no significant performance variation, whereas "/" indicates that no relevant study or corresponding result is currently available in the literature.

The remainder of this paper is organized as follows: Section II presents an overview of high-quality open-source datasets prevalent in AD perception tasks, accompanied by a preliminary analysis. Sections III and IV delve into the perception algorithms utilized in radar and camera technologies, respectively, highlighting recent developments and comparing the merits and drawbacks of various approaches. Section V examines the typical stages of multi-sensor fusion, detailing the techniques employed in various fusion methods and organizing the perception tasks categorically. Sections VI and VII engage in an in-depth discussion on the methodological differences, evaluating the strengths and limitations of current methods. They further provide an analysis of the present state of the field and identifies ongoing challenges and obstacles within multi-sensor fusion for AD perception, aiming to provide researchers with insights into potential future research directions and issues warranting further investigation.

2. Open-source Datasets for AD

The development of AD agent systems and the optimization of perception algorithms heavily rely on high-quality datasets. With the advent of transformative artificial intelligence technologies such as deep learning and large language models, a substantial number of real-world datasets and simulations that closely mimic real-world conditions have been introduced in recent years (Alibeigi et al., 2023; Caesar et al., 2020; Cong et al., 2022; Geiger et al., 2012; Geyer et al., 2020; Huang et al., 2018; Mao et al., 2021; Sun et al., 2020; T. Sun et al., 2022; Wilson et al., 2023; Xu et al., 2023; Yu et al., 2020; Yu et al., 2022; Shi et al., 2026). The quality of these datasets greatly impacts the ability of AD systems to maintain safe, reliable, and efficient operations in complex and variable driving scenarios. Table 2 lists the representative open-source datasets commonly used in AD perception tasks and presents a detailed description of their characteristics. It is evident that nearly all datasets focus on collecting multiple sensor modalities across diverse scenarios, including variations in weather and lighting conditions, and most datasets support a wide range of perception tasks. This diversity significantly facilitates the development of multi-sensor fusion algorithms. In the following part, five representative datasets

Table 2
Representative Open-source Datasets and Benchmarks for Autonomous Driving.

Dataset	Year	Cameras	LiDAR	Tasks					Scenarios	Diversity
				2D detection	3D detection	Segmentation	Tracking	Lane Detection		
KITTI (Geiger et al., 2012)	2012	2*Color and 2*grayscale PointGrey Flea2 Video Cameras (1392*512)	1*Velodyne HDL-64E 3D Laser Scanner (64 beams)	√	√				Urban, Suburban, Highway	/
ApolloScape (Huang et al., 2018)	2018	2*Cameras (3384*2710)	2*VUX-1HA Lser Scanners	√	√	√	√	√	Urban, Suburban, Highway	Area, Light
nuScenes (Caesar et al., 2020)	2020	6*RGB Cameras (1600*900)	1*Spinning LiDAR (32 beams) 5*MMWR (77GHz)	√	√	√	√	√	Urban, Suburban	Weather, Area, Light
Waymo (Sun et al., 2020)	2020	5*Cameras (1920*1280 of F/FL/FR; 1920*1040 of SL/SR)	5*LiDARs (75m of TOP; 20m of F/SL/SR/R)	√	√	√	√		Urban, Suburban	Weather, Area, Light
BDD100K (Yu et al., 2020)	2020	RGB Camera(s) (1080*720)	/	√		√	√	√	Urban, Highway	Weather, Area, Light
A2D2 (Geyer et al., 2020) (Audi Autonomous Driving Dataset)	2020	6*RGB Cameras(2.3MP)	5*LiDAR (16 beams)		√	√			Urban, Suburban Highway	Weather, Area
ONCE (Mao et al., 2021)	2021	7*Cameras (1920*1920)	1*LiDAR (40 beams)		√				Urban, Suburban, Highway, Tunnel	Weather, Area, Light
STCrowd (Cong et al., 2022)	2022	1*Monocular Camera	LiDAR (128 beams)	√	√		√		Urban	Weather, Area, Light
SHIFT (T. Sun et al., 2022)	2022	5*RGB Cameras (1280*800), 1*Stereo RGB Camera Set, A Depth Camera	1*LiDAR (128 beams)	√	√	√	√		Urban, Suburban	Weather, Area, Light
DAIR-V2X (Yu et al., 2022)	2022	1*Infrastructure RGB Camera (1920*1080), 1*Vehicle RGB Camera (1920*1080)	1*Infrastructure LiDAR (300 beams), 1*Vehicle LiDAR (40 beams)		√				Urban	Weather, Area, Light
ZOD (Alibeigi et al., 2023) (Zenseact Open Dataset)	2023	1*Camera (3848*2168)	1*Velodyne VLS128 Rotating 3D Laser Scanner, 2*Velodyne VLP16	√	√	√		√	Urban, Suburban	Weather, Area, Light
AV2 (Wilson et al., 2023) (Argoverse 2)	2023	7*Global shutter Cameras	2*LiDAR (32 beams)		√		√		Urban	Weather
V2V4Real (Xu et al., 2023)	2023	1*RGB Camera (1280*720) on Tesla, 1*RGB Camera (2064*1544) on Ford	1*LiDAR (32 beams)		√		√		Urban, Highway	Weather, Area

will be selected for a brief overview analysis, highlighting their specific features and the contributions they offer to the field of AD research.

nuScenes. In the realm of AD, the nuScenes (Caesar et al., 2020) dataset represents a landmark benchmark, employing an extensive sensor array that encompasses six RGB cameras, one LiDAR, and five MMWRs. This assembly has facilitated the collection of 1,000 richly annotated driving scenes for AD research. Predominantly, the data encompass urban and rural scenes within Boston and Singapore. It is noteworthy that this dataset pioneers the inclusion of diverse lighting conditions and intricate weather scenarios, such as nocturnal and rainy environments, significantly augmenting the dataset's diversity and complexity relative to contemporaneous datasets. Consequently, this dataset serves as a robust platform for the optimization and validation of models pertaining to prevalent tasks in the field, including but not limited to object detection, segmentation and tracking, and lane detection.

DAIR-V2X. DAIR-V2X (Yu et al., 2022) represents a landmark dataset resource, integrating infrastructure-side and vehicle-side sensor data. Released in 2022, the dataset distinguishes itself through the deployment of two distinct sensor arrays: infrastructure sensors and onboard vehicle sensors. This configuration enabled the comprehensive large-scale multimodal data collection of urban driving scenarios in Beijing. Primarily, the dataset facilitates research into AD issues pertaining to vehicle-infrastructure cooperative perception, with a particular focus on the realm of 3D object detection.

ApolloScape. ApolloScape (Huang et al., 2018), a dataset released by Baidu in 2018, represents a significant advancement in the field of AD. Baidu, a company with a commercial AD division, has curated a dataset that provides comprehensive support for research in the major perception tasks of AD, including 2D/3D object detection, semantic segmentation, object tracking, and lane detection. The dataset's data collection apparatus comprises two laser scanners and two front-facing cameras, which have been employed to gather and annotate images and point clouds of Beijing's intricate traffic scenes and highways. Additionally, acknowledging the diversity of lighting conditions, the dataset categorizes and organizes data from different periods of the day, ensuring comprehensive coverage of varying illumination intensities.

Waymo. The Waymo dataset (Sun et al., 2020) was released at the IEEE Conference on Computer Vision and Pattern Recognition in 2020, encompassing more than 1000 traffic scenes each containing 20-second sequences, and covering approximately 76 square kilometers of complex urban and suburban environments in the United States. Data collection spanned various weather conditions, offering a robust benchmark for research tasks covering geographically and meteorologically diverse conditions. The deployment of five LiDAR units and cameras at varied locations and angles ensures the dataset's high quality and large scale. This dataset provides an excellent benchmark and robust support for tasks such as detection, segmentation, and multi-object tracking.

ONCE. The ONCE dataset (Mao et al., 2021), released by Huawei in 2021, represents a large-scale autonomous driving dataset in the AD landscape. As a company that also boasts an autonomous vehicle development division equipped with AD components, Huawei amassed approximately one million LiDAR scenes and seven million corresponding RGB images over 144 hours of driving, approximately twenty times the volume of the nuScenes dataset. Utilizing seven cameras and a 40-beam LiDAR, the dataset encompasses a broad spectrum of urban, suburban, highway, and tunnel regions across different times and weather conditions, meticulously collected and annotated. Aimed at setting a benchmark for large-scale 3D perception research, the dataset has undergone extensive evaluation and analysis through various self-supervised and semi-supervised methods, affirming its high quality.

Overall, existing AD datasets exhibit significant differences in scenario coverage, environmental diversity, sensor configurations, and supported perception tasks. Datasets such as nuScenes and Waymo mainly emphasize complex urban and suburban driving environments, while ApolloScape focuses more on intricate traffic scenes and highway scenarios under varying illumination conditions. In contrast, DAIR-V2X specifically highlights vehicle–infrastructure cooperative perception through the integration of infrastructure-side and vehicle-side sensing systems. Moreover, datasets such as nuScenes and ONCE further improve environmental diversity by incorporating different weather conditions, nighttime scenes, tunnels, and large-scale traffic scenarios. From the perspective of sensor diversity, recent datasets increasingly integrate cameras, LiDAR, MMWR, and infrastructure-side sensing devices to support multi-modal perception and fusion research. These differences indicate that current AD datasets are gradually evolving toward larger-scale, more diverse, and more complex real-world driving scenarios, thereby providing important support for the development and evaluation of robust multi-sensor fusion algorithms (Liu et al., 2025; Wang et al., 2026).

3. Camera-based Perception for AD

In the dynamic landscape of AD, cameras play an indispensable role in the perception system of vehicles, serving as critical sensors that provide rich visual information essential for environment perception and scene understanding. As the eyes of autonomous systems, cameras capture detailed environmental information, allowing vehicles to perceive and interact with the world around them. This capability is fundamental not only for basic navigation tasks such as lane keeping and obstacle avoidance, but also for complex decision-making processes that enable high-level autonomous driving functions (Dong et al., 2025; Vinayak et al., 2026).

Cameras are one of the main sensors used in self-driving cars as they provide high-resolution visual data and rich semantic information. Their effective sensing distance typically ranges from a few tens of meters to more than one hundred meters, but the robustness can be greatly affected by lighting conditions and weather conditions (Badue et al., 2021). The effectiveness of these camera systems hinges on the sophistication of the underlying algorithms that analyze and interpret the captured imagery. This analysis spans a variety of approaches, each tailored to enhance the ability of the system to detect, classify, and respond to diverse scenarios encountered on the road. Firstly, we delve into the importance of innovations in machine vision algorithms for AD and explore the evolution from basic feature recognition to advanced deep learning strategies, underscoring the transformative impact these developments have had on the field of machine vision. Secondly, we highlight the critical role of cameras in AV perception,

detailing their essential contributions to 3D tracking, short-range perception, lane detection, and traffic sign recognition, which collectively enhance vehicle safety and navigational capabilities.

3.1. Computer Vision-based Algorithms for AD

As the primary sensors in numerous systems, especially in AVs, cameras utilize a range of algorithms to process and interpret the visual data they capture. Unlike the point cloud data collected by MMWR and LiDAR, which provide 3D spatial information, the image data captured by cameras offer rich texture and color information, presenting unique challenges and opportunities for processing and interpretation in the context of AD systems. These algorithms can be broadly categorized into three main approaches: traditional methods, machine learning-based methods, and deep learning-based methods.

3.1.1. Traditional Methods on Camera

Traditional video analysis techniques have predominantly relied on fundamental feature extraction methods to interpret and understand visual content effectively. Historically, these initial approaches concentrated on manually designed, handcrafted features extracted directly from raw pixel data. Serving as the cornerstone for more advanced analytical methods, these traditional techniques have facilitated the development of innovative approaches in video analysis. This is particularly evident in applications such as vehicle detection, where precision and reliability are crucial. Typically, these traditional methods are categorized into three main types: symmetry-based, edge-based, and color-based, each playing a vital role in enhancing the accuracy and efficiency of video analysis systems.

Symmetry-based methods. Symmetry-based methods leverage the inherent geometric symmetry typical of vehicles, particularly when viewed from the front or rear, to effectively differentiate them from asymmetrical background elements (Alessandretti et al., 2007; Chunmei et al., 2016; Gern et al., 2001; Satzoda and Trivedi, 2015; Teoh and Bräunl, 2012). Algorithms such as summed normalized cross-correlation (Nishigaki et al., 2012) are often utilized to evaluate symmetry by comparing mirrored segments of an image around a central axis. This approach is instrumental in distinguishing objects from background clutter, enhancing the clarity and accuracy of the analysis. However, despite their practicality, symmetry-based methods encounter challenges such as sensitivity to environmental factors, including lighting variations, which can distort the apparent symmetry of a vehicle and lead to false negatives. Furthermore, these methods are computationally intensive, particularly when processing high-resolution images. Their dependency on clear symmetrical views restricts their effectiveness to specific angles, thereby limiting the applicability of symmetry-based detection in diverse observational scenarios.

Edge-based methods. Edge-based methods in vehicle detection exploit the distinct linear boundaries of vehicles, employing algorithms such as Canny (Woo et al., 2017) and Sobel (Srinivasa et al., 2003) to enhance these edges within images. These methods are particularly valued for their computational efficiency, which renders them suitable for real-time applications in AVs in early real-time systems. By identifying sharp changes in image brightness, edge detection algorithms delineate the horizontal and vertical lines that define vehicle structures, such as windows and bumpers (Mu et al., 2016; Nur et al., 2016; Sole et al., 2004; Srinivasa, 2002). However, these methods may struggle with background textures that resemble vehicle edges, potentially resulting in false positives.

Color-based methods. Color-based methods in vehicle detection utilize the distinct color patterns of vehicles to differentiate them from their surroundings. This approach leverages the specific color channels within digital images, applying threshold segmentation techniques to isolate the colors characteristic of vehicles from the background (Wang et al., 2011; Wang et al., 2016). By discerning color of the vehicle from a typically less colorful environment, such as an asphalt road, this method can effectively highlight vehicle presence. However, color-based detection is notably sensitive to environmental variables that affect visual perception (Aytekin and Altuğ, 2010; Chen et al., 2015; Russell and Zou, 2012). Illumination changes due to weather conditions, time of day, or shadows can alter the perceived color of a vehicle, complicating the segmentation process and potentially leading to inaccuracies in vehicle detection.

3.1.2. Machine Learning-based Methods on Camera

The advent of machine learning (ML) has catalyzed a transformative shift in the field of automated driving, significantly enhancing the accuracy and efficiency of these systems. At the heart of these advancements are sophisticated feature extraction techniques that adeptly differentiate vehicles from their surroundings. Key among these techniques are Haar-like features and Histogram of Oriented Gradients (HOG), which serve as crucial elements within ML algorithms. Fig. 1 shows the camera perception process based on machine learning.

Haar-like features. Haar-like features, initially developed for facial recognition (Viola and Jones, 2001), have been adeptly repurposed for vehicle detection (Wen et al., 2014). They utilize 2D nonstandard Haar wavelets across vertical, horizontal, and diagonal orientations to capture essential structural elements like lines, edges, and surrounding geometry of objects (Papageorgiou et al., 1998). Despite their computational simplicity, these features often require additional processing to manage the high dimensionality of their feature vectors (Wang et al., 2015). To address this, alternative robust methods such as Gabor features (Ming and Jo, 2011) and Speeded Up Robust Features (Bay et al., 2006) are explored to boost the performance of detection systems in varied environments.

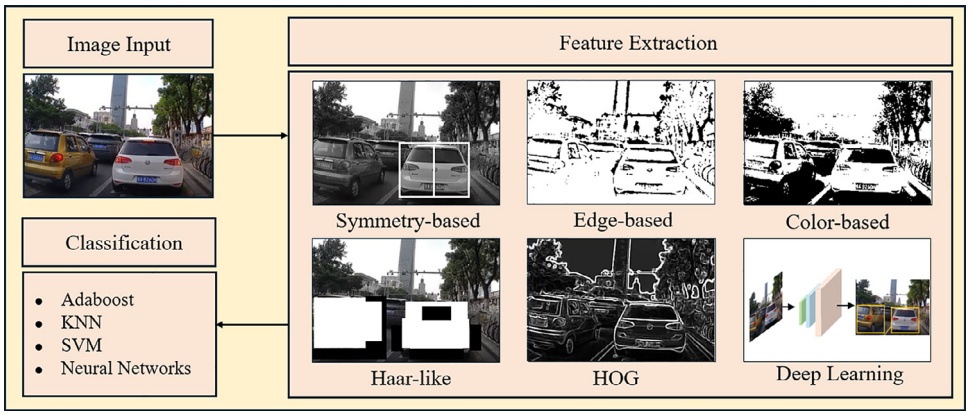


Fig. 1. General process of machine learning-based methods on camera.

HOG. HOG features have proven widely used in object detection, especially in scenarios involving pedestrians and vehicles. Their resilience against lighting variations and partial occlusions makes them particularly valuable. HOG works by calculating statistical histograms of gradient directions in localized image sections, capturing critical edge and silhouette information. This capability has been expanded through adaptations that integrate multiple HOG descriptors or configurations, enhancing the ability to detect more complex patterns (Chavez-Garcia and Aycard, 2015; Khairdoost et al., 2013; Otto et al., 2012; Yan et al., 2016).

ML Models for Classification. Furthermore, an essential aspect of ML models in vehicle detection involves classifying images into early vehicle and non-vehicle classification tasks using the aforementioned feature extraction methods (Wen et al., 2014). This classification is performed by training classifiers such as KNN (Ali et al., 2020), SVM (Cheon et al., 2012; Hsieh et al., 2014) and AdaBoost (Chen and Meng, 2013.; Sivaraman and Trivedi, 2014), which are adept at handling high variability within classes and establishing discriminative decision boundaries through statistical learning. The integration of multiple feature classifiers has demonstrably improved detection outcomes. Additionally, to mitigate the computational demands of processing entire images, a focused approach has been adopted that directs ML classifiers to analyze only Regions of Interest (RoI) (Yan et al., 2016). This strategy not only enhances the efficiency but also significantly improves the real-time performance of detection systems, demonstrating the substantial impact of ML in advancing vehicle detection technologies.

3.1.3. Deep Learning-based Methods on Camera

The rapid advancement of computing power and the emergence of extensive datasets for training have catalyzed remarkable progress in deep learning. By harnessing the capabilities of large-scale datasets and employing intricate neural network architectures, deep learning models have gained the ability to derive hierarchical representations of raw visual data, thereby identifying complex patterns and subtle details that were previously challenging to capture. This evolution marks a significant departure from traditional handcrafted feature extraction techniques, resulting in a profound transformation of camera-based perception systems within AD (Tang et al., 2023). Such a paradigm shift has driven notable advancements in critical AD tasks, including object detection, semantic/instance segmentation, and object tracking.

Object Detection. Traditional object detection methods (Solunke and Gengaje, 2023) typically relied on a combination of sliding-window strategies (Lampert et al., 2008), feature extraction (Wang, 2011), and classification techniques (Ragland and Tharcis, 2014). These approaches involved systematically scanning an image at various positions and scales to determine whether each candidate window contained the target object (Gualdi et al., 2012). In recent years, an increasing number of researchers have designed deep learning neural networks specifically for object detection tasks, which are typically categorized into one-stage detectors and two-stage detectors (Kaur and Singh, 2024).

One-Stage Detectors. Single-shot detectors, also referred to as one-stage detectors, perform object detection in a single forward pass during inference, outputting bounding boxes and class probabilities. This single-step framework directly maps image pixels to bounding box coordinates and class probabilities, thereby reducing computational complexity and allowing for real-time performance (Zou et al., 2023). Examples of single-stage detectors include You Only Look Once (YOLO) (Redmon et al., 2016), Single-Shot Multibox Detector (SSD) (Liu et al., 2016), RetinaNet (Li and Ren, 2019), EfficientDet (Tan et al., 2020), CornerNet (Law and Deng, 2018), CenterNet (Duan et al., 2019), Detection Transformer (DETR) (Carion et al., 2020), and Swin Transformer (Liu et al., 2021). These methods are renowned for their efficiency and capability to operate in real-time, making them particularly suitable for applications requiring rapid detection.

Table 3
Evolution of the YOLO Series for Object Detection.

Version	Key Architectural Innovation	Primary Strengths	Noted Limitations / Focus	Reference
YOLOv1	**Unified CNN for regression-based detection.**	**Real-time speed, simplified pipeline.**	**Poor small object detection, localization errors.**	(Redmon et al., 2016)
YOLOv2	**Darknet-19 backbone; introduced batch normalization and anchor boxes.**	**Improved recall and small object detection.**	**–**	(Redmon and Farhadi, 2017)
YOLOv3	**Darknet-53 backbone; multi-scale prediction with feature pyramid networks.**	**Better accuracy across object scales, handles more classes.**	**–**	(Redmon and Farhadi, 2018a)
YOLOv4	**CSP-DarkNet53 backbone; PANet for feature fusion; SPP module; Mish activation.**	**State-of-the-art balance of accuracy and inference speed.**	**Increased architectural complexity.**	(Bochkovskiy et al., 2020)
YOLOv5	**Modified CSP-DarkNet53 backbone; integrated AutoAnchor for anchor optimization.**	**Reduced memory/computational cost; user-friendly training pipeline.**	**–**	(Zhang et al., 2022)
YOLOv6	**EfficientRep backbone; optimized for industrial deployment.**	**High inference speed and hardware efficiency.**	**Focused on deployment rather than novel architecture.**	(Li et al., 2022)
YOLOv7	**Extended Efficient Layer Aggregation Networks (E-ELAN); reparameterized convolutions.**	**New state-of-the-art speed-accuracy balance.**	**–**	(Wang et al., 2023)
YOLOv8	**Anchor-free detection head; streamlined architecture with integrated NMS.**	**Simplified pipeline, efficient training and inference.**	**–**	(Varghese and Sambath, 2024)
YOLOv9	**Generalized Efficient Layer Aggregation Network (GELAN); Programmable Gradient Information (PGI).**	**Enhanced accuracy while maintaining lightweight design.**	**–**	(Wang et al., 2025)
YOLOv10	**NMS-free, end-to-end design; optimized model components.**	**Real-time end-to-end object detection.**	**–**	(Wang et al., 2024)
YOLOv11	**Enhanced YOLOv8 architecture with C3k2 and C2PSA modules; extended task support.**	**Improved detection accuracy; supports pose estimation and instance segmentation.**	**Very recent, community evaluation ongoing.**	(Alif, 2024; Khanam and Hussain, 2024)
YOLOv12	**Attention-centric architecture within the YOLO framework.**	**Enhanced detection accuracy while maintaining real-time inference speed.**	**Very recent, limited independent validation.**	(Tian **et al., 2025)**

YOLO. YOLO (You Only Look Once) was proposed in 2015, introducing a regression-based object detection paradigm that utilizes a single CNN architecture to directly predict bounding boxes and class probabilities from an entire image (Redmon et al., 2016). This unified framework enabled real-time detection by processing images in a single forward pass, representing a significant shift from earlier multi-stage detectors. However, early versions exhibited limitations in precisely localizing small objects and generalizing across diverse scales. Subsequent iterations have systematically addressed these challenges through continuous architectural innovation, solidifying YOLO's role as a foundational model for real-time perception in autonomous driving. Table 3 summarizes the key developments across the YOLO family.

The evolution outlined in Table 3 reveals several consistent trends: (1) progressive backbone refinement from Darknet variants to more efficient architectures like CSP-DarkNet53 and GELAN, enhancing feature extraction; (2) advancements in multi-scale feature fusion through mechanisms such as feature pyramids, PANet, and adaptive spatial pooling (SPP) to handle objects of varying sizes; and (3) a continuous drive to optimize the speed-accuracy trade-off, culminating in recent end-to-end, anchor-free designs that simplify the detection pipeline while maintaining competitive performance. Despite these advances, challenges remain for automotive applications, including robustness under adverse weather, occlusion handling, and efficient deployment on edge hardware—areas where YOLO-based research continues to evolve.

SSD. SSD streamlined the detection process by integrating object localization and classification into a single CNN pass, eliminating the need for region proposal stages found in earlier models like Faster Region-based Convolutional Neural Networks (Faster-RCNN). The SSD architecture was based on a VGG-16 backbone and was followed by progressively smaller additional convolutional layers. It employed multi-scale feature maps and predefined default boxes with varying aspect ratios and scales to detect objects of different sizes within an image (Liu et al., 2016). SSD combined the regression idea of YOLO and the anchor mechanism of Faster-RCNN and used the multi-scale region at each position of the whole map for regression, which not only maintains the speed of YOLO, but also ensures that the prediction is accurate like Faster-RCNN (Ganga et al., 2024).

Feature Fusion SSD (FSSD) used a lightweight feature fusion module that concatenated features from various layers and generated a framework for pyramidal feature maps, thus only trading only a small speed loss for higher accuracy (Li et al., 2024). DF-SSD replaced the original VGG-16 backbone network with a more advanced network DenseNet and obtained better feature extraction results. Its advantages in enhancing feature propagation and reducing the vanishing gradient problem brought a higher detection

accuracy (Zhai et al., 2020). DBQ-SSD introduced a dynamic ball query network that adaptively selected a subset of input points based on the feature information, significantly reducing the computational cost while maintaining the detection accuracy (Yang et al., 2023). FCR-SSD transferred the shallow semantic information to deeper layer through the cross-enhanced feature fusion module and the attention mechanism, which improved the feature representation of small objects and achieved more accurate object detection (Gong et al., 2023).

Transformer. The Transformer architecture was proposed by Vaswani et al. in 2017. It has created significant interest in the field of deep learning by introducing a self-attention mechanism to enable models to capture long-range dependencies (Vaswani et al., 2023a). Its original design was for natural language processing tasks, but its ability to model complex relationships has been widely used in the field of computer vision tasks, particularly for object detection.

DETR redefined the object detection task as a direct set prediction problem, eliminating the need for components such as non-maximum suppression and anchor generation (Carion et al., 2020). Its architecture comprised a CNN (typically ResNet-50 or ResNet-101) for feature extraction and was followed by a Transformer encoder-decoder. The encoder processed the feature maps into embedding sequences, while the decoder used learned object queries to predict object bounding boxes and class labels. This end-to-end approach simplifies the detection pipeline and achieves competitive performance on the benchmark datasets. Deformable attention modules were implemented in Deformable DETR to address slow convergence and poor small object accuracy in DETR (Xia et al., 2022; Zhu et al., 2021). These modules allowed the model to focus on a set of relevant features to improve efficiency. The architecture retained the Transformer encoder-decoder structure but replaced standard attention with deformable attention, which provided the advantage of dealing with objects of different sizes and has faster training speed. Swin Transformer used a hierarchical vision Transformer (ViT) that computed representations with shifted windows to capture both local and global context (Deshmukh et al., 2023; Liu et al., 2021). This design allowed Swin Transformer to serve as a generalized backbone network with other detection heads like Faster R-CNN (Ren et al., 2017) or Mask R-CNN (He et al., 2017) for a variety of visual tasks. Vision Detection Transformer (ViDT) introduced a reconfigured attention module to expand the Swin Transformer as a standalone detector (Song et al., 2021). This was followed by a Transformer decoder that exploited multi-scale features and achieved a new balance between computational cost and accuracy.

Two-Stage Detectors. These approaches typically involve two stages: the generation of candidate regions and the subsequent classification and bounding box regression for these regions. In other words, they first generate a set of potential RoIs and then perform precise classification and localization on these regions. This strategy of filtering followed by detailed recognition often results in higher detection accuracy, but at the cost of increased computational overhead and slower processing speeds (Kaur and Singh, 2024). Common basic examples of two-stage detectors include R-CNN (Girshick et al., 2014), Fast R-CNN (Girshick, 2015), Faster R-CNN (Ren et al., 2017), FPNs (Feature Pyramid Networks) (Lin et al., 2017), and Mask R-CNN (He et al., 2017).

R-CNN was a groundbreaking model proposed by Girshick et al. in 2014 (Girshick et al., 2014). It combined region proposals with deep learning for object detection. The process started with generating approximately 2,000 region proposals with selective search, and then each proposal was resized and fed into the CNN to extract the features. These features were then classified by using an SVM (Support Vector Machine), and bounding box regressors were implemented to refine the object locations. While R-CNN achieved significant accuracy, it was computationally inefficient due to the need to process each region proposal separately. To address this problem, Girshick implemented Spatial Pyramid Pooling Network (SPPNet) and proposed Fast R-CNN in 2015 (Girshick, 2015). This approach used VGG16 to perform feature extraction on entire images and combined classification and regression steps to improve the performance. In 2015, Faster R-CNN (Ren et al., 2017) was proposed based on previous works. It integrated the region proposal generation into the network itself. The key innovation was the fully convolutional network (FCN) Region Proposal Network (RPN) that predicted object bounds and objectness scores at each position. The RPN shared the convolutional layer with the detection network with almost cost-free region proposals (Zhang et al., 2017). The FPN (Feature Pyramid Network) addressed the multi-scale challenges in object detection tasks (Lin et al., 2017). It fully used the hierarchical feature pyramid in CNNs to significantly improve the detection performance for small objects with minimal additional computation cost. In 2017, He et al. extended the framework of Faster R-CNN and proposed Mask R-CNN (He et al., 2017). It was able to complete both object detection and instance segmentation tasks by adding an extra branch in parallel with the existing one for classification and bounding box regression. One of the key innovations of Fast R-CNN was the implementation of the RoIAlign layer, which addressed the problem of the original RoI pooling by preserving the exact spatial locations. RecFRCN proposed a novel method to improve the bounding box regression accuracy by recalibrating the categories of these boxes (Zhang and Lu, 2023). The method integrated a new classification network called Rec-Net into the Box Predictor of Faster R-CNN, which contained a feature extractor, a Feature Enhancement Block (FEB), a RoI pooling layer and local classifier. The FEB was designed to enhance the extracted features and thus improve the ability to distinguish between object classes. Object detectors in autonomous driving are commonly categorized by their prediction pipeline. The following table summarizes their typical accuracy–efficiency trade-offs and deployment considerations.

(Table 4)

Segmentation. Image segmentation is fundamental to pattern recognition and image understanding, which involves dividing an image into meaningful regions. Over the years, segmentation techniques have advanced from simple approaches to deep learning architectures, significantly improving their applicability and accuracy in various fields. Early segmentation methods mainly relied on simple algorithms or mathematical theories, such as thresholding (Lee et al., 1990), edge detection (Muthukrishnan and Radha, 2011), clustering-based (Coleman and Andrews, 1979), random walks (Grady, 2006), and graph theory approaches (Peng et al., 2013).

Table 4
Comparison of Representative Object Detectors for Autonomous Driving.

Model family	Representative methods	Strengths in AD	Limitations in AD	Typical use
One-stage CNN detectors	YOLO, SSD, RetinaNet, EfficientDet	Low latency. Simple pipeline. Good throughput on embedded hardware.	Small or distant objects can be harder. Accuracy can drop under heavy occlusion.	Real-time camera perception on the vehicle.
Two-stage CNN detectors	Faster R-CNN, FPN, Mask R-CNN	Strong localization. Higher recall for small objects. More stable accuracy under clutter.	Higher compute and memory. Slower inference. Less suitable for strict real-time budgets.	Accuracy-oriented modules, offline analysis, or high-end hardware stacks.
Transformer-based detectors	DETR, Deformable DETR, Swin-based detectors, ViDT	Global context reasoning. Flexible modelling of long-range relations. Can reduce hand-crafted components.	Higher compute cost. Training can be slower and data-hungry.	Research prototypes, high-capacity platforms, or hybrid backbones in fusion stacks

Thresholding converted a grayscale image to a binary image by selecting a threshold of global or local intensity and classified pixels above that threshold as foreground and those below as background (Kohler, 1981). Edge detection techniques, including different operators such as sobelX, sobelY (Vincent and Folorunso, 2009), Prewitt (Yang et al., 2011), and Kirsch (Hui et al., 2011), identified boundaries within images by detecting discontinuities within the images to highlight object contours. Clustering-based methods (Dubey, 2018; Mittal et al., 2022) clustered pixels with similar features together using a matrix in the feature space after mapping the image to the feature space. Region-based methods (region growing and region merging) (Kaganami and Beiji, 2009; Karthick et al., 2014; Girshick et al., 2016) segmented an image based on pixel intensity or texture uniformity within a region. Random walk approaches distinguished between different objects by assigning labels to neighboring elements according to predefined rules (Grady, 2006; Meilă and Shi, 2000; Dong et al., 2016). The methods based on graph theory converted images into graphs (Peng et al., 2013; Basavaprasad and Hegadi, 2017). They used related theories to analyze the graph with weights and find the optimal segmentation by computing the global optimization of the graph. The Markov Random Field (MRF) (Anguelov et al., 2005; Wu and Chung, 2007; Kato and Zerubia, 2012; Liu et al., 2018; Wang and Ma, 2024) modeled the spatial dependencies of pixel labels and implemented probabilistic segmentation that included contextual information.

ML-based methods introduced a probabilistic model that considered the spatial relationship between pixels. In addition, with the development of deep learning, the ability of models to learn complex features directly from data revolutionized this field (Minaee et al., 2022). Fully Convolutional Networks (FCN) (Long et al., 2015) replaced fully connected layers with convolutional layers. Specifically, its core idea was to replace the fully connected layers in the traditional CNN AlexNet and VGGNet with fully convolutional layers to build a pixel-level segmentation network. It enabled frame-by-frame pixel prediction and end-to-end training and laid the foundation for the application of deep learning in segmentation tasks (Ajmal et al., 2018; Huang et al., 2022).

SegNet (Badrinarayanan et al., 2017) was an encoder-decoder structure with pooled indices to preserve spatial information. U-Net (Ronneberger et al., 2015) was an encoder-decoder structure introduced by Ronneberger et al. for biomedical image segmentation and used skip connections to further improve localization accuracy and capture contextual information. This design helped to fully synthesize the underlying and higher-level information of the image and hence was widely used in image processing in the medical field (Siddique et al., 2021). For feature fusion, U-Net adopted an innovative strategy by performing feature concatenation and fusion along the channel dimension rather than in a pointwise fusion approach used in FCN. This method enabled better preservation and utilization of feature information so that the performance had been improved as a result.

DeepLab was a segmentation model developed by Google's research team based on CNNs (Padthe et al., 2024). The DeepLab family has evolved through four major versions: V1, V2, V3, and V3+. DeepLabV1 (Chen et al., 2016) was an improvement of Deep Convolutional Neural Networks (DCNN) to address the two major problems in segmentation tasks. To address the resolution degradation and detail loss caused by repeated pooling and downsampling operations, DeepLabV1 introduced Atrous Convolution. This technique increased the receptive field while retaining more positional information. In addition, to enhance segmentation accuracy and details, DeepLabV1 employed a fully connected Conditional Random Field (CRF), which improved the capture of local features and delineation of details such as image contours by integrating pixel positions, values of surrounding pixels, and global contextual information. DeepLabV2 (Chen et al., 2018) further improved on V1 by introducing the Atrous Spatial Pyramid Pooling (ASPP) module for the challenge of multi-scale object segmentation. It resampled and fused features to better capture information at different scales. It retained the use of atrous convolution and fully connected CRF to further improve detailed information. DeepLabV3 (Chen et al., 2017) made important improvements to V2 by dropping the fully connected CRF and instead emphasizing deeper atrous convolutional structures, and improving the ASPP module. By concatenating atrous convolutions with different expansion rates, DeepLabV3 was able to capture image features at different scales, enabling the network to better understand structural and semantic information at different scales. DeepLabV3+ (Chen et al., 2018) built on V3 by maintaining the exclusion of fully connected CRFs and introducing Depthwise Separable Convolution as an encoder and added a Decoder module to the model. The design combined the strength of Spatial Pyramid Pooling (SPP) (He et al., 2015), which acquired more scale information, with an encoder-decoder structure that recovered fine boundary details. This integration allows DeepLabV3+ to achieve superior performance in both capturing semantic context and restoring object boundaries.

Mask R-CNN (He et al., 2017) extended Faster R-CNN with a masking hierarchy for predictive detail masking, which can perform the segmentation task by further recognizing its contours in addition to the object detection task. The feature pyramid network

extracted high-resolution and semantically rich features accommodating objects with significant size variations. However, the computational complexity of Mask R-CNN was high and therefore required high hardware resources. During both training and inference, the model must handle a large number of convolutional and deconvolutional operations, as well as the cropping and alignment of multiple candidate regions, which significantly increase computational and storage demands (Shu et al., 2020).

Recently, the success of Transformers in natural language processing has motivated researchers to explore its applications in visual tasks (Vaswani et al., 2023a). Transformer's unique encoder-decoder architecture allowed them to effectively handle uncertain inputs and capture long-range dependencies, and various segmentation models based on it were gradually available at the state-of-the-art performance on related datasets (Strudel et al., 2021; Yao et al., 2024). Early improvements combined self-attention layers with convolutional layers to enhance CNNs, while subsequent research attempted to use pure self-attention mechanisms instead of convolutional layers (Wang et al., 2023). Since then, ViTs (Dosovitskiy et al., 2021a) directly employed sequences of image patches to classify entire images. Meanwhile, DETR (Carion et al., 2020) introduced the concept of object queries. It replaced the complex anchor-based designs of earlier frameworks and simplified the process of detection and segmentation. Numerous follow-up studies were improved upon them (Khan et al., 2022; Thisanke et al., 2023). For example, SegFormer (Xie et al., 2024) first employed a hierarchical backbone to extract features with different resolutions like ResNet, and improved performance and reduced the number of parameters through finer-grained segmentation, eliminated positional encodings, and other designs (Li et al., 2022; Bai et al., 2023).

As a new generation of deep learning models, Visual Foundation Models (VFMs) (Yang et al., 2024; Yu et al., 2024) can effectively capture rich visual features by training on massive data. MetaAI proposed Segment Anything Model (SAM) (Kirillov et al., 2023; Ravi et al., 2024) for image segmentation. Similar to GPT-4 (Achiam et al., 2024) in natural language processing, SAM was a prompt-based model to adapt to images across diverse imaging modalities, which provided new possibilities for the field of segmentation (Hetang et al., 2024; Huang et al., 2024; Huang et al., 2024; Zhang et al., 2024). For instance, Zhang et al. proposed the medical SAM Adapter (Med-SA) (Wang et al., 2024), which employed lightweight but effective adaptation techniques to incorporate domain-specific medical knowledge into the segmentation models. It introduced two innovative components: Spatial-Depth Transposition (SD-Trans) for adapting 2D SAM to 3D medical images, and the Hyper-Prompt Adapter (HyP-Adpt) for prompt-based adaptation. Med-SA outperformed several SOTA medical image segmentation methods with only 2% parameter updates.

Object Tracking. Object tracking is one of the popular topics in computer vision. It usually refers to constructing a robust model to predict the motion states such as shape, size, and position of the objects in the video after setting the target of interest in the initial frame (Zhang et al., 2024). Early object tracking methods were primarily dominated by traditional methods, including MeanShift (Zhou et al., 2009), Particle Filters (Jaward et al., 2006), Kalman Filters (Gunjal et al., 2018), and optical flow methods (Kale et al., 2015). Despite their relatively low complexity, these methods were often unable to adapt to complex environments and handle challenges such as occlusion. This limitation was mainly due to their inadequate feature representation, the lack of background consideration, and limited tracking capabilities regarding motion trajectories (Hassan et al., 2024). With the rapid development of deep learning, researchers have since explored various models for tracking tasks, leading to significant progress in this field.

Early algorithms usually focused on object-based modeling to identify corresponding features of an object by modeling its appearance (Luo et al., 2021). Examples of such approaches included region matching, feature point tracking, active contour-based, and optical flow (Hassan et al., 2024). Additionally, some researchers have attempted to incorporate prediction algorithms such as Kalman Filter, Particle Filter, MOSSE (Han, 2017), KCF (Maharani et al., 2023), SAMF (Liu et al., 2023), BACF (Kiani Galoogahi et al., 2017), etc. into the tracking task. These algorithms may also combine strategies such as MeanShift (Zhou et al., 2009) or CamShift (Nouar et al., 2006) to reduce the search scope.

Later, researchers retained the general framework of these traditional tracking methods but replaced handcrafted features like HOG, grayscale, CN, and Harris features with those extracted from deep learning methods (Cao et al., 2024). The Deep Learning Tracker (DLT) (Wang and Yeung, 2013) method used a multilayer autoencoder network for feature generation, directly applying deep learning techniques to the tracking tasks. It addressed the problem of missing positive samples during online training and proposed a method that combined offline training and offline tuning. To improve the feature characterization ability of the fully connected based network, Wang et al. proposed an algorithm for learning hierarchical features that was able to adjust the pre-learned features online with the corresponding module (Wang et al., 2015). In 2015, Ma et al. proposed a visual tracking algorithm based on hierarchical convolutional features (HCF) that was based on the KCF network (Ma et al., 2015). Its ability to retain high resolution features and preserve semantic information significantly improved object classification effectiveness. In 2016, Qi et al. introduced HDT, which adopted an ensemble learning approach to train filters using six different feature layers individually (Qi et al., 2016). Through the HEDGE algorithm, it achieved self-adaptive weight adjustment and improved tracking performance. The DeepLMCF (Wang et al., 2017) method enhanced tracking accuracy and robustness in complex situations by incorporating deep features into the original LMCF model. Since the front layer contained more location information, DeepSRDCF (Danelljan et al., 2015) built a single-layer network on top of SRDCF and used deep features extracted by VGGNet, which not only reduced the computational complexity but significantly improved the performance. In 2017, Danelljan et al. introduced the ECO tracker (Danelljan et al., 2017) based on the C-COT (Danelljan et al., 2016) method. ECO improved tracking by refining quality filters, optimizing the sample sets through Gaussian mixture model and adopting a sparse model update strategy, which mitigated the effects of occlusions and reduced update frequency.

To efficiently extract and update the deep features of an object in real time, offline end-to-end trained twin network-based tracking methods have gained significant attention, owing to their faster tracking speeds and higher accuracy. Bertinetto et al. (2016) developed an end-to-end Siamese network, SiamFC, to frame the tracking problem as a similarity learning task. The interconnection between

two inputs was computed through a bilinear layer, with shared weights between branches, which enabled efficient sliding-window evaluations. In 2018, Li et al. proposed the SiamRPN tracking method (Li et al., 2018), a twin RPN, which integrated the RPNs of Faster R-CNN (Ren et al., 2017) into the SiamFC framework, leveraging feature fusion. However, treating tracking as a classification problem makes it susceptible to interference from similar objects. SANet (Fan and Ling, 2017) employed recurrent neural networks (RNNs) to create a structure-aware network that extracts the target's structural information, enhancing both the ability to distinguish between similar objects and to separate the target from the background. Track R-CNN (Shuai et al., 2020; Voigtlaender et al., 2020; Xie et al., 2021), by contrast, applied 3D convolution within the Mask R-CNN framework to synthesize contextual information, allowing feature maps to pass through the RPN to accomplish the tracking task.

The applications of transformers have evolved beyond natural language processing (NLP) and are now being applied to various computer vision tasks because of their strong ability to model global interactions and long-range dependencies. Through self-attention, cross-attention, positional coding and other methods, it can realize overall perception and macroscopic understanding of image sequences. In tracking tasks, it is often used to augment and fuse features extracted by CNNs (Meinhardt et al., 2022). In 2021, Wang et al. introduced transformers to the tracking domain, which used attention mechanisms to establish pixel associations between video frames and augment features by conveying richer temporal information (Wang et al., 2021). Additionally, the authors improved the structure of the Transformer to facilitate high-quality feature generation. The performance of both frameworks is further improved by combining the twin network tracking framework SiamFC with the discriminative filter TrDiMP. MOTR (Zeng et al., 2022) extended the object detection end-to-end network DETR to the MOT domain by proposing the Track Query method in combination with the track label-aware strategy Tracklet-Aware Label Assignment (TALA) to realize object tracking. As the first fully end-to-end MOT model based on the Transformer architecture, MOTR achieved implicit target association and implemented iterative trajectory prediction through frame-by-frame updating, effectively learning temporal variations in the data. SwinTrack (Lin et al., 2024), built upon the Swin-Transformer architecture, performed feature extraction and fusion. Within the Siamese framework (Bromley et al., 1993), it embedded and enhanced the temporal context of historical trajectory information through the integration of motion tokens, facilitating efficient feature matching and tracking while processing both the target object and search region.

Furthermore, the application scope of deep learning algorithms transcends traditional visual data processing and is progressively being extended to broader camera perception domains. Most notably, by incorporating advanced perception modalities such as fisheye cameras and event cameras at the frontend of camera systems, deep learning techniques can further harness the unique advantages of these modalities.

Fisheye camera is a kind of special camera equipment with an ultra-wide viewing angle, usually with a field of view of 180 degrees or even higher, and is often used to realize wide-area environment perception and surround-view monitoring in autonomous driving. Its significant advantage is the ability to reduce blind zones, which makes it particularly suitable for applications such as automated parking, narrow road navigation, and surround view stitching. However, because fisheye lenses introduce significant image distortion and resolution non-uniformity, traditional perception algorithms may suffer performance impacts when processing fisheye images. In recent years, with the development of deep learning technology, researchers have proposed a variety of algorithms for fisheye aberration correction, feature extraction, and perceptual optimization, making the role of fisheye cameras in autonomous driving perception systems increasingly prominent. Fisheye cameras have a relatively small angular resolution despite their large angular volume coverage, so they are unable to detect smaller objects at long distances and are generally used as near-range sensors. In a typical Surround View Camera (SVC) system, a configuration of four fisheye cameras is commonly used to cover the entire 360° near-field area around the vehicle (Kumar et al., 2023).

As an emerging visual sensor, event cameras have received extensive attention in the field of AD for their unique working principle. Unlike traditional cameras that record the scene frame by frame, event cameras record light intensity change events at the pixel level to realize sparse and efficient data acquisition. The amount of data in the event stream is much smaller than that transmitted by conventional cameras, and the event stream has no minimum time unit, so it has low latency characteristics unlike conventional cameras that output data at regular intervals. Event cameras capture brightness changes asynchronously via an event-driven approach, typically reporting timestamps with subtle-level time resolution and high dynamic range, making them well suited for high-speed, dynamic scenes (Shariff et al., 2024). This feature provides event cameras with an ultra-high dynamic range and microsecond response speed, which is especially suitable for dynamic target detection, motion estimation, and perception tasks in complex lighting environments. However, the unstructured form of data generated by event cameras poses a new challenge to traditional algorithms, and deep learning models based on event data are being developed for target detection and tracking.

3.2. The Application of Camera in AD Perception

Cameras play a pivotal role as the eyes of the vehicle, providing essential visual information that supports numerous safety and navigation functions. These sophisticated devices capture and analyze vast amounts of data to interpret road signs (Marti et al., 2019; Wen and Jo, 2022; Yeong et al., 2021), detect pedestrians (Marti et al., 2019), recognize traffic lights (Marti et al., 2019; Wen and Jo, 2022; Yeong et al., 2021), and monitor other vehicles (Yeong et al., 2021) with remarkable accuracy. The integration of camera technology in AVs transcends simple image capture; it includes complex processing algorithms that enable vehicles to make informed decisions in real time (Theivadas and Ponnan, 2026). As autonomous systems continue to evolve, the reliance on camera technology not only enhances the capability of these systems to interact safely with their environment but also drives forward advanced developments in AV technologies.

3.2.1. 3D Perception

Stereo vision markedly enhances 3D perception applications by leveraging dual cameras to produce dense depth maps through the calculation of disparities between two aligned images (Hamzah and Ibrahim, 2016). This technique, akin to biological vision, offers critical spatial data necessary for precise object tracking. Unlike other depth-sensing technologies such as LiDAR, which require high contrast or textured surfaces to function effectively, stereo vision can generate comprehensive depth information in a variety of settings. This capability makes it particularly valuable in contexts like AV navigation and robotic systems, where environmental complexity demands robust sensing solutions (Broggi et al., 2015). When integrated with platforms such as the Robotic Operating System (ROS) (Harapanahalli et al., 2019), stereo vision facilitates advanced functionalities like simultaneous localization and mapping (SLAM). This integration enhances the interaction and navigation capabilities of autonomous systems, ensuring reliable and accurate 3D perception across diverse and dynamic environments, thus bolstering the operational efficacy of these technologies.

3.2.2. Near-field Perception

Fisheye cameras have become increasingly vital in the context of AVs, particularly for near-field sensing tasks such as parking assistance and maneuvering through traffic jams (Heng et al., 2019; O'Mahony et al., 2018; Yogamani et al., 2019). Their wide-angle lenses offer a comprehensive 360-degree view using just four cameras, a setup that dramatically enhances the spatial awareness vehicle. These cameras are typically integrated with convolutional neural network (CNN) architectures, significantly boosting the capability for object detection and segmentation in the vicinity of the vehicle. For instance, certain implementations have been demonstrated to operate at 15 frames per second while achieving a mean Intersection over Union (IoU) accuracy of 69.5%, showcasing robust real-time performance in dynamic environments (Yahiaoui et al., 2019). This advanced technological integration not only reinforces the safety features of AVs but also markedly enhances their operational efficiency in crowded spaces. As such, fisheye cameras have become an essential element in the toolkit of modern AD systems, contributing significantly to their adaptability and safety in complex driving scenarios.

3.2.3. Lane Detection

Lane detection is an essential component of advanced driver assistance systems (ADAS) and AD technologies. Historically, this process depended on handcrafted features to determine the position and shape of road lanes. However, as technology has advanced, deep learning methods have largely supplanted these traditional techniques, providing a more dynamic and robust approach to lane recognition. Contemporary deep learning strategies primarily employ semantic segmentation, where each pixel is classified to ascertain whether it belongs to a lane. This method, typically implemented using CNNs (Bittel et al., 2017; Lyu et al., 2019) combined with recurrent neural networks (RNNs) (Zou et al., 2019), facilitates accurate and continuous lane detection under diverse conditions.

Lane detection methodologies can be categorized into three main types based on their operational focus: Ego-lane detection, Ego-road Lane detection, and All-roads Lane detection. Ego-lane detection concentrates on identifying the current lane and its boundaries, which is crucial for maintaining the position of vehicle within its lane (Chen and Chen, 2017; Liu et al., 2017; Oliveira et al., 2016). Ego-road lane detection expands upon this by recognizing additional road attributes such as lane numbers and types of lane markings, aiding in lane changes and navigation decisions (Hou et al., 2019; Lee et al., 2017; Li et al., 2016; Pan et al., 2018). All-roads lane detection aims for a comprehensive identification of all visible road markings, including those on opposite lanes, and is primarily used in offline applications such as high definition map modeling for advanced navigation systems (Azimi et al., 2018; Bai et al., 2018; Garnett et al., 2019; Kurz et al., 2019). This structured approach to lane detection ensures that vehicles operate safely and efficiently within varied driving environments.

3.2.4. Traffic Sign Detection and Recognition (TSDR)

TSDR are indispensable elements within the framework of AD systems, necessitating the incorporation of advanced vision techniques to ensure safe navigation. These systems predominantly employ two methods for the identification and classification of traffic signs: color-based and shape-based approaches. Color-based methods exploit the distinct, vivid colors of traffic sign to distinguish them from their backgrounds (Liu et al., 2016; Luo et al., 2017). Research has shown that transformations in color spaces (Wen and Jo, 2017), from RGB to YUV for instance, significantly enhance the accuracy of traffic sign classification. Additionally, techniques such as adaptive histogram equalization are utilized to standardize color representation, thereby improving detection robustness under various lighting condition (Møgelmose et al., 2015).

On the other hand, shape-based methods utilize the geometric properties of signs to aid in recognition. Advanced segmentation tasks that integrate both color and shape information have been developed to refine the detection process (Zhu et al., 2017). Moreover, enhancements to existing deep learning architectures, such as incorporating additional regression vectors into models like Faster R-CNN (Lee and Kim, 2018), have been effective in improving the estimation of traffic sign boundaries, thus boosting detection accuracy. These methodologies underscore the complexity and sophistication required in TSDR systems to maintain safety and efficiency in AV operations.

4. Depth-sensing System-based Perception for AD

The advancement in AD technology has elevated the importance of depth sensors within vehicle perception systems. These sensors play a pivotal role in detecting objects by emitting laser or electromagnetic waves and capturing their reflections, thus providing real-time information about the distance, speed, and orientation of target objects (Zhu et al., 2022). Common depth sensors deployed on AVs for perception tasks include LiDAR, MMWR, and Ultrasonic radar. Point cloud data consists of discrete points in three-dimensional

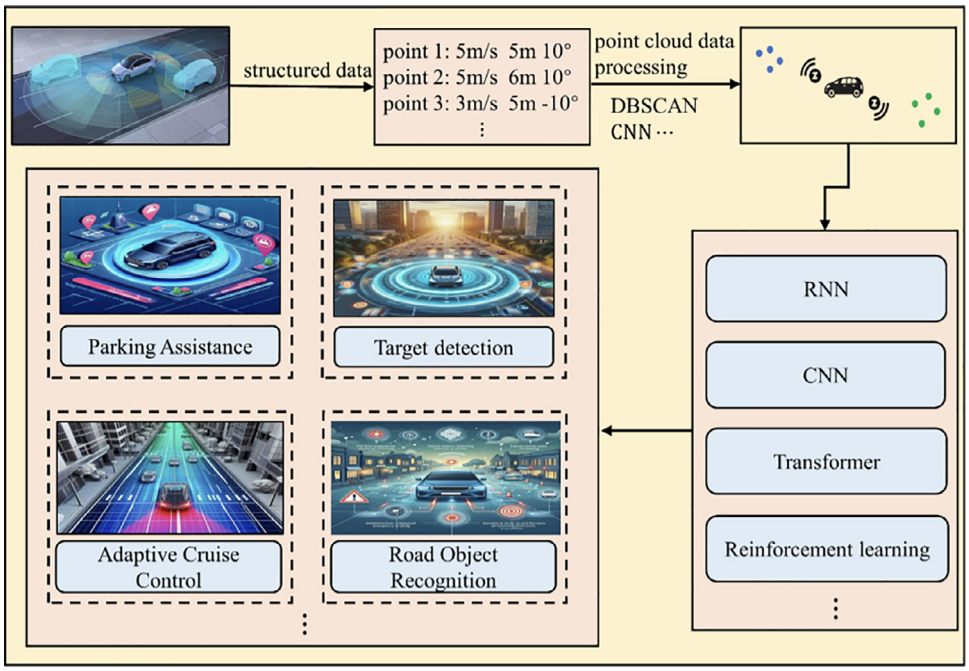


Fig. 2. Approaches to Radar in AD.

space, where each point typically contains coordinate information (x, y, z) and attributes such as reflectance intensity. Since point cloud data is inherently unstructured and lacks the regular 2D grid arrangement of images, it generally requires specific representation or encoding for perception in AVs to facilitate feature extraction and related operations. In deep learning models, after data from depth-sensing systems undergoes a cleaning and denoising process similar to that of RGB data, it must then be subjected to special data representation conversions and feature extraction steps. The most fundamental representation is the raw point cloud, consisting directly of point coordinates and attributes like reflectance intensity, as output by the sensor. This representation preserves the original information to a great extent since no projection or similar operations are applied, thus avoiding information loss. However, due to the unstructured nature of raw point clouds, they are difficult for conventional neural networks to process directly.

In the early stages of 3D object detection and semantic segmentation, methods such as PointNet was employed to process these raw point clouds (Charles et al., 2017). For depth sensing systems, point cloud data is most commonly converted into voxelized representations (Saglam et al., 2020). This representation divides the three-dimensional space into a series of fixed-size voxels. Statistical operations (such as point density, average reflectance intensity, and coordinate-based statistical features) or feature encoding are performed on the points within each voxel. This approach transforms the original unstructured point cloud into a structured, grid-like 3D data format. Its advantage lies in providing a regular 3D structure, allowing the direct application of 3D convolutions or sparse convolutions on the three-dimensional grid for feature extraction. In 3D space, this representation effectively leverages spatial neighborhood information, helping the model achieve better local feature extraction and multi-scale analysis. For example, VoxelNeXt utilizes voxelized representations to design networks for precise 3D object detection and tracking tasks (Chen et al., 2023). Additionally, many voxelization-based approaches that integrate classic deep learning models have become a primary direction in point cloud research for various perception tasks, such as Voxel R-CNN (Deng et al., 2021). Another special 2D rasterization method is known as Pillarization. This method considers only information on the horizontal plane (X-Y plane) while storing Z-direction information within the features. That is, the space is divided into vertical pillars, and points within each pillar are statistically processed or encoded to form a 2D feature map. A similar approach commonly used in multimodal fusion is the Bird's Eye View (BEV) representation. Specific details will be discussed in the following section. The following subsections detail the principles, functionalities, and limitations of the two most widely used radars. Fig. 2 illustrates the common radar-based AD perception algorithm framework.

In summary, although camera-based perception methods have achieved remarkable progress in object detection, segmentation, and tracking tasks, their performance still heavily depends on environmental conditions such as illumination, weather, and occlusion. Moreover, monocular or RGB-based vision systems inherently lack accurate depth perception and geometric spatial representation capabilities, which limits their robustness and reliability in complex autonomous driving scenarios. Therefore, relying solely on camera sensors remains insufficient for robust AD perception under diverse real-world driving environments.

4.1. Working Principle

4.1.1. MMWR

MMWR is a radar system that operates within the millimeter wave band (typically between 30 GHz and 300 GHz) to detect and gather information about targets, such as their distance, speed, and angle. Millimeter-wave radar can consistently detect the relative

distance and velocity of objects at ranges of hundreds of meters and remains highly resistant to interference in varying environmental extremes (Zhou et al., 2020).

Typically, vehicle-mounted MMWR operates in Frequency Modulated Continuous Wave (FMCW) mode (Gao et al., 2021; He et al., 2023). FMCW radar determines target speed and distance by assessing the frequency difference between the transmitted and received signals. Offering higher frequency and shorter wavelength than traditional microwave radar, MMWR provides enhanced resolution and accuracy, making it well-suited for target detection and tracking in complex environments, particularly in AD and drone applications (Cai et al., 2021; Zhang et al., 2023). With the ongoing advancements in RF device integration, especially radar-on-chip and RF integrated circuits, MMWR can achieve high-resolution point cloud imaging within the limited space of a vehicle. Point cloud, comprising all data points acquired by MMWR via electromagnetic wave reflection, facilitates target tracking and positioning through array signal processing and deep learning technologies (Cheng et al., 2022; Gao et al., 2021; Zhang et al., 2023).

Automotive MMWR has mainly used 24 GHz and the 76–81 GHz band (often referred to as 77/79 GHz). Legacy 24 GHz short-range radars typically provide limited bandwidth and thus coarser range resolution, and the industry has largely shifted toward 76–81 GHz solutions as regulations and hardware ecosystems evolved. Frequency choice affects both resolution and practical use. The 76–81 GHz band supports wider sweep bandwidth, enabling finer range resolution and better separation of nearby objects in dense traffic. As a result, these radars contribute not only robust range and Doppler measurements, but also improved spatial evidence that can strengthen multi-sensor fusion, particularly under low visibility where cameras and LiDAR may degrade (Zhan et al., 2024a), especially in for, rain, snow, or low illumination, depending on sensor type and waves.

Automotive radar is robust under low illumination and many adverse weather conditions, and it directly measures radial velocity via the Doppler effect. Its main limitation lies in spatial detail. Radar returns are typically sparse and have limited angular resolution, which makes fine-grained shape recovery difficult and reduces the ability to separate closely spaced objects in dense traffic. Radar measurements can also be affected by multi-path reflections and interference, which may introduce false targets and reduce reliability in complex urban environments (Yang et al., 2022; Zhan et al., 2024b).

4.1.2. LiDAR

The operational steps of LiDAR are similar to those of MMWR, but it employs lasers instead of electromagnetic waves. LiDAR can provide high-precision 3D point cloud data, and its sensing range is usually around two hundred meters. Despite its precise spatial measurement capabilities, these devices are typically more expensive and performance degrades significantly in extreme weather and when encountering objects with low reflectivity (Zhao et al., 2024).

These lasers, usually emitted by low-power near-infrared lasers at wavelengths between 900 nm and 1050 nm, are invisible to the naked eye. Analogous to MMWR, LiDAR generates a point cloud representation of the scene based on laser beam reflections. When the beam illuminates an object, it reflects, enabling precise 3D spatial positioning. Various processing methods are utilized for LiDAR point clouds (Cui et al., 2022), including 2D Views Representation (Klokov and Lempitsky, 2017), Volumetric Representation (Li et al., 2020; Maturana and Scherer, 2015; Qi et al., 2016), Point Representation (Lan et al., 2019; Ma et al., 2021), Index/Tree Representation (Zeng and Gevers, 2019), Graph Representation (Chen et al., 2020; Defferrard et al., 2016).

Automotive LiDAR can be divided into mechanical, hybrid solid-state, and all-solid-state designs based on the scanning mechanism. Mechanical LiDAR is still a mature choice for high fidelity 3D perception because it can achieve extensive coverage and dense point clouds through rotating components. Its practical disadvantages include increased size, expense, and long-term reliability issues. Solid-state LiDAR replaces macroscopic rotation with electronic or micro-actuated beam steering, including MEMS scanning, flash LiDAR, and optical phased arrays. These designs are more compact and meet the requirements of automotive packaging and mass production. However, the solid-state system still faces engineering trade-offs, such as limited field of view, beam-steering efficiency and performance stability under actual driving conditions (Li et al., 2022).

LiDAR provides high-precision 3D geometry and accurate ranging, supporting lane-level localization and detailed obstacle reconstruction. Its performance, however, can degrade in rain, fog, and snow due to scattering and attenuation, which can lead to missing or noisy returns. Cost and packaging constraints remain practical barriers as well, particularly for mechanical scanning units (Kim et al., 2023).

4.2. The Application of Radar in AD Perception

4.2.1. MMWR

Object Detection and Tracking. MMWR facilitates the detection and monitoring of various targets such as vehicles, pedestrians, and obstacles on the road (Chang et al., 2020; Zhang et al., 2023; Zhang et al., 2022), thereby enhancing the real-time environment sensing capabilities of AD systems.

Road Object Recognition. Through the recognition of diverse objects on the road, including vehicles, traffic signs (Dang et al., 2023; Dong et al., 2023), and road boundaries (Grebner et al., 2023; Wijesoma et al., 2004), MMWR aids in enhancing the AD system's comprehension of the surrounding environment, leading to more precise driving decisions.

Environment Sensing. MMWR assists in perceiving different environmental conditions surrounding the vehicle, such as road and weather conditions (Ge et al., 2018; Wijesoma et al., 2004), enabling adaptive adjustments to driving strategies and speeds.

Adaptive Cruise Control. Leveraging the detection capabilities of MMWR for vehicles and traffic conditions ahead, AVs can achieve adaptive cruise control (Abou-Jaoude, 2003; Tsang et al., 2006; Yuan et al., 2018; Zhang et al., 2020), dynamically adjusting speeds while maintaining safe distances.

Collision Prevention and Emergency Braking. MMWR supports anticipating potential collision risks (He et al., 2021), enabling AVs to use emergency braking systems to prevent accidents proactively.

Lane Keeping and Lane Change Assist. By monitoring the vehicle's lane position, MMWR assists drivers in maintaining lanes and executing lane changes (Yuan et al., 2018), enhancing overall driving safety.

Parking Assistance. MMWR aids AVs in executing parking maneuvers (Cai et al., 2021; He et al., 2021; Lin et al., 2021), including parallel and perpendicular parking, thereby improving parking accuracy and safety.

4.2.2. LiDAR

There are numerous applications of LiDAR in AD vehicles, as shown below.

Obstacle Detection and Tracking. LiDAR accurately detects and tracks obstacles on the road (Asvadi et al., 2016; Wu et al., 2023; Yuan et al., 2022), such as vehicles, pedestrians, and other objects, thereby assisting the AD system in real-time perception of the surrounding environment.

Mapping and Localization. LiDAR can generate high-precision environmental maps and provide precise vehicle localization (Elghazaly et al., 2023; Hata and Wolf, 2016), helping AVs accurately perceive and understand the surrounding roads and terrain.

Road Recognition and Boundary Detection. By identifying various road signs, lane markings, and boundaries, LiDAR helps the AD system better recognize roads and maintain proper lane positioning (Han et al., 2012; Horváth et al., 2022).

Adaptive Cruise Control. Based on LiDAR's detection of vehicles and traffic conditions ahead, AVs can achieve adaptive cruise control (Davis, 2021), automatically adjusting speed and maintaining a safe distance.

Collision Prevention and Emergency Braking. LiDAR helps AVs predict potential collision risks and trigger emergency braking systems to avoid collisions (Srinivasa, 2002; Srinivasa et al., 2003).

Lane Keeping and Lane Change Assistance. By monitoring the vehicle's lane position, LiDAR assists drivers in maintaining the vehicle within the correct lane and executing lane changes when necessary (Kim et al., 2023).

Parking Assistance. LiDAR assists AVs in parking operations, including parallel and perpendicular parking, thereby improving parking accuracy and safety (Imameev et al., 2021; Lakshmanan et al., 2023).

Overall, compared with camera-based perception systems, LiDAR and MMWR sensors provide stronger geometric perception capability, depth awareness, and environmental robustness under challenging conditions. Nevertheless, each sensing modality still exhibits inherent limitations in terms of semantic understanding, sensing resolution, environmental adaptability, or deployment cost. Therefore, although these sensing technologies have significantly improved autonomous driving perception performance, relying solely on a single sensing modality remains insufficient for achieving robust and reliable perception across diverse and complex driving environments.

5. Multi-sensor Fusion

The limitations of single-modality perception systems discussed in Sections 3 and 4 have motivated extensive research on multi-sensor fusion for autonomous driving perception. By integrating heterogeneous sensing information from cameras, LiDAR, MMWR, and other sensing devices, multi-sensor fusion methods can effectively combine complementary semantic, geometric, and motion features, thereby improving perception accuracy, robustness, and environmental adaptability under complex driving conditions. Consequently, an increasing number of studies have focused on multi-sensor fusion frameworks, particularly those combining LiDAR and cameras (Gu et al., 2021; Li et al., 2022; Liu et al., 2022; Liu et al., 2023; Pang et al., 2022; Zhou et al., 2023), or MMWR and cameras (Chadwick et al., 2019; Dong et al., 2021; Lin et al., 2024; Schramm et al., 2024; Zhou et al., 2023), have increasingly been considered by researchers. This section will summarize the common multi-modal fusion methods currently used in the field and elaborate on them in detail, categorizing them according to their technical orientations and the specific perception tasks they target. These discussions aim to illustrate how these integrated approaches enhance the accuracy and reliability of perception systems in AVs by leveraging the complementary strengths of different sensor modalities.

5.1. Introduction to Multi-sensor Fusion

Perception tasks in AD are typically categorized into two main perspectives: macroscopic motion information and microscopic road-environment information perception. Common sensors equipped on AVs include LiDAR, cameras, MMWR, ultrasonic radar, inertial sensors, and GPS, among others (Feng et al., 2021). From a purely functional perspective, autonomous agents require a diverse array of sensors due to the varying functionalities of each sensor type, which support robust perception and safe autonomous operation. From a demand perspective, real-world driving scenarios exhibit complex diversity such as varying lighting conditions and weather states, making the perception methods reliant on single or multiple individual sensors lack robust generalization capability (Bijelic et al., 2020; Dong et al., 2021; Feng et al., 2021; Han et al., 2024; Rubio et al., 2013; Zhang et al., 2021). Furthermore, from an industrial standpoint, multi-sensor fusion is considerably more cost-effective under the same perceptual performance conditions compared to employing multiple sensors of a single type (Zhao et al., 2024; Zhou et al., 2023; Pang et al., 2022; Dong et al., 2021; Pang et al., 2020). Therefore, theoretically, the performance of multi-sensor fusion methods surpasses that of any individual sensor perception performance (Wang, 2021). Given the strengths and weaknesses of sensors like cameras, LiDAR, and MMWR deployed on AVs, efficiently leveraging complementary information from each component presents a major challenge in multi-sensor fusion.

In recent years, the use of bird's-eye view (BEV) spatial representations has become an important innovation in the field of autonomous driving perception. BEV has also become a fundamental basis for addressing challenges related to traditional sensor modalities in the sensor fusion process (Man et al., 2023; Singh, 2023; Zhao et al., 2024). In the two types of sensor modality data representations primarily focused on in this review, traditional camera-based methods operate within 2D perspective views. This approach effectively captures rich semantic information, such as object appearance, color, and texture. However, monocular or stereo images inherently suffer from ambiguity and uncertainty in depth information, especially in complex urban environments with varying lighting conditions, distant and small objects, and occlusions. In contrast, LiDAR and MMWR operate in 3D space. The 3D point cloud space contains abundant and relatively precise spatial and depth information, which can maintain good performance even under adverse weather conditions, such as heavy fog or rain (Ma et al., 2024). However, such data is highly dependent on sensor sparsity and lacks the ability to comprehensively understand real-world scenes due to the absence of dense semantic details. In this context, BEV effectively mitigates many of these limitations by transforming data from multiple sensor modalities into a unified spatial framework. Unlike the original sensor data, BEV allows for the merging of heterogeneous data sources (cameras, LiDAR, and MMWR) into homogeneous 3D planar data. This operation achieves cross-modal alignment in spatial dimensions (Fu et al., 2024). Specifically, this transformation operation first extracts features from RGB image data obtained from cameras and then projects the extracted features onto the BEV space by combining the camera's intrinsic and extrinsic calibration matrices with depth estimation algorithms. In this transformation process, whether using a monocular camera or sensor fusion methods, depth estimation is a critical step for accurately reconstructing three-dimensional world coordinates. Furthermore, in the 3D space, the vertical axis (z-axis) of the world coordinate system can be directly discarded to achieve the flattening process for obtaining BEV projections. This is also the primary approach currently employed by many methods to address point cloud perception challenges through the BEV space. LiDAR and radar data are converted into BEV planes by compressing their vertical dimensions, ensuring spatial alignment of each modality on the 2D plane. This process effectively overcomes the limitations of individual sensors, providing a unified spatial understanding and making it easier to combine and interpret sensor data from different sources. After data from different sensors are mapped to the BEV space, the fusion of features from these different modalities becomes a core advantage of the BEV representation. The spatial consistency provided by BEV makes it the main framework for multi-sensor fusion, ensuring integrity and robustness for ADAS (Li et al., 2024; Shi et al., 2024). For example, in the BEV space, fusing camera features with LiDAR data allows the integration of rich semantic information from cameras with precise geometric data from LiDAR. In this process, the BEV space unifies the overhead planes of the two types of information. Meanwhile, pixel-level or region-level comparison and fusion operations enhance the spatial interaction granularity of the data from these two modalities. This approach significantly improves the model fusion accuracy. More specifically, this collaboration addresses common challenges such as depth ambiguity in cameras and enhances the ability to detect and track targets, particularly those that are distant or partially occluded. Similarly, radar data provides basic velocity information, complementing the static spatial representation of LiDAR and the dense textures of cameras, thereby facilitating the accurate detection and motion estimation of dynamic targets (Zhao et al., 2024). Moreover, for ADAS applied to complex traffic scenarios, fusion efficiency must be considered alongside fusion accuracy when designing algorithm models. Compared to traditional multimodal fusion methods, fusion in the BEV space avoids the information redundancy and error propagation associated with converting features from each modality into a unified 3D coordinate space. Additionally, since BEV representation maps features from different modalities to a two-dimensional plane, it supports various types of sensor combinations. For complete AVs, features in the BEV space are also more easily integrated with other modality information, such as high-precision maps, to achieve high-accuracy perception and decision-making. Furthermore, in recent years, due to the exceptional performance of Transformers in various computer vision tasks, attention mechanisms have also come into researchers' focus. BEV can similarly utilize attention-based fusion techniques, using BEV grid queries to selectively focus on sensor-specific features (Lin et al., 2024). Specifically, BEV features can directly query relevant features from each modality through attention modules. These features encompass key information from different modalities, such as semantic information from cameras, spatial geometric information from LiDAR, and velocity information from radar. The self-attention mechanism is capable of dynamically balancing the contributions of each modality's features and adaptively selecting different methods for information combination.

In the design of fusion strategies, the initial consideration is the stage of fusion. The chosen stage of fusion dictates the overall structure of the algorithmic framework. Commonly, fusion stages are categorized into early fusion and late fusion (Wang et al., 2023).

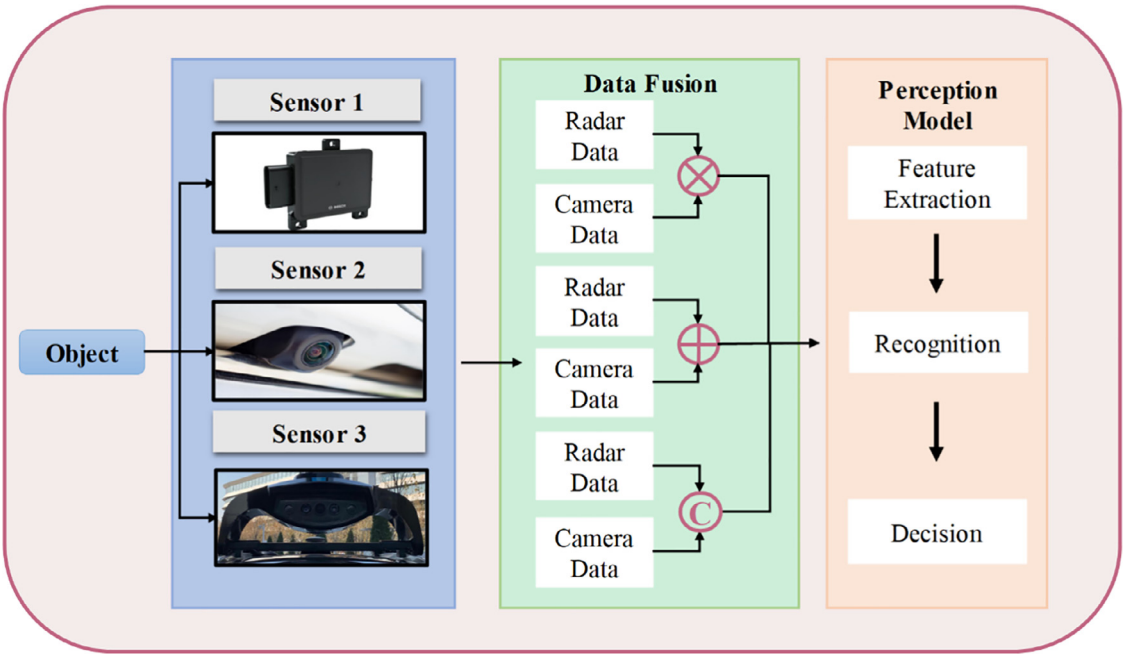


Fig. 3. Demonstration of the data fusion architecture.

5.1.1. Early Fusion

Early fusion in the context of perception tasks for AD refers to the method of integrating various modal data or multi-modal feature extraction at the preliminary stages of processing. This fusion process occurs prior to the prediction phase in different branches. The primary advantage of this fusion approach is the direct interaction enabled between different modal data or features at the intermediate layers. As the fusion takes place during the early to mid-stages of the entire perception framework, it allows for maximal utilization of features including semantic and depth information. However, a significant challenge accompanying early fusion is the utilization of diverse modal information, particularly the issues of feature representation within a single modality and the spatial-temporal alignment across multiple modalities.

Commonly, early fusion in AD perception tasks is differentiated into data-level fusion and feature-level fusion (Feng et al., 2021). Data-level fusion involves combining different types of data collected by various sensors by projecting one type of data into the space of another, followed by subsequent feature extraction and predictive decision-making processes as illustrated in Fig. 3. Because operations at the data level preserve the integrity of the original information, the results are typically the most precise. However, data-level fusion requires stringent spatial- temporal alignment of raw data collected from multiple sensors and is computationally intensive, which imposes a significant burden on system communication. It also demands homogeneity among sensors, i.e., that multiple sensors observe the same physical phenomenon. Thus, only a few studies utilize data-level fusion exclusively (Alfred Daniel et al., 2023). Another approach is feature-level fusion. This process begins after different sensors have collected data. Each set of data undergoes feature extraction. Subsequently, feature maps or vectors from different spaces are integrated, as depicted in Fig. 4.

The simplest form of feature-level fusion is the utilization of depth information from point cloud data as a new dimension for RGB image features, integrated via channel concatenation, thus forming RGB-D information. This stacked RGB-D feature map is then used for subsequent object detection or semantic segmentation tasks (Chadwick et al., 2019). Additionally, some studies employ a cascading method, combining features extracted from LiDAR point cloud data with RGB images (Liu et al., 2022), which helps address issues such as detection inaccuracies due to the sparsity of point clouds.

5.1.2. Late Fusion

Late fusion is often referred to as target-level or decision-level fusion in the context of AD perception (Huang et al., 2022). Decision-level fusion involves first performing perception tasks on data from different sensors using separate branches, and then integrating the detection results obtained from these branches, as shown in Fig. 5.

The essence of this methodology lies in establishing correspondences among prediction results from different types of data to enable matching. Traditional approaches, for example, involve calculating similarities based on the position and size of the detected objects to facilitate matching (Fadadu et al., 2022). Additionally, metrics such as the IoU are also used as indicators for data association (Fu et al., 2024). Techniques like Kalman filtering and DeepSORT, which are frequently employed in tracking and target matching, are also commonly used in target-level multi-modal fusion.

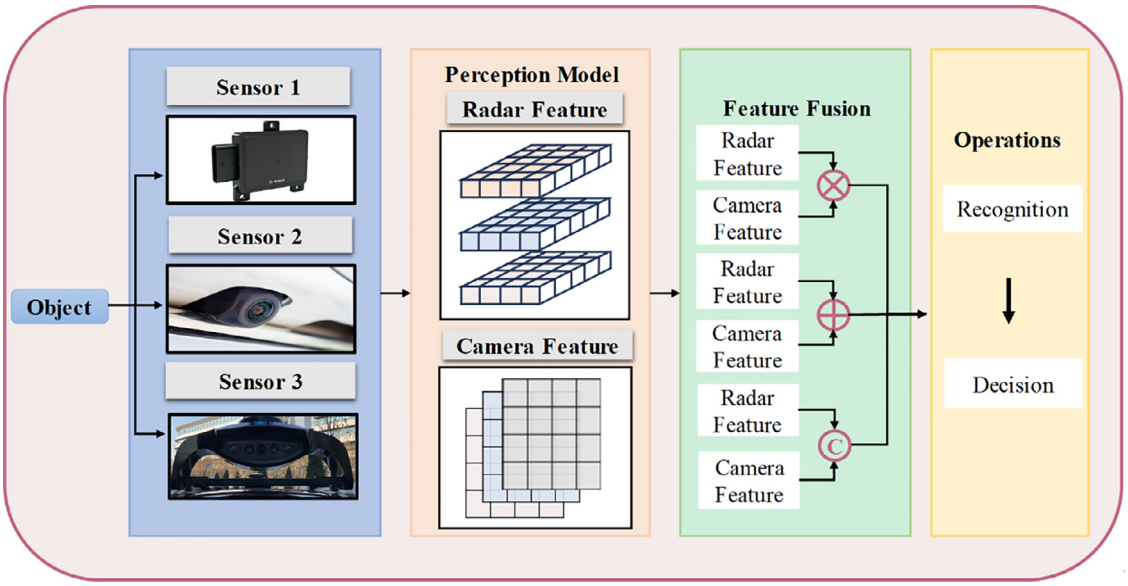


Fig. 4. Schematic representation of the feature fusion frame.

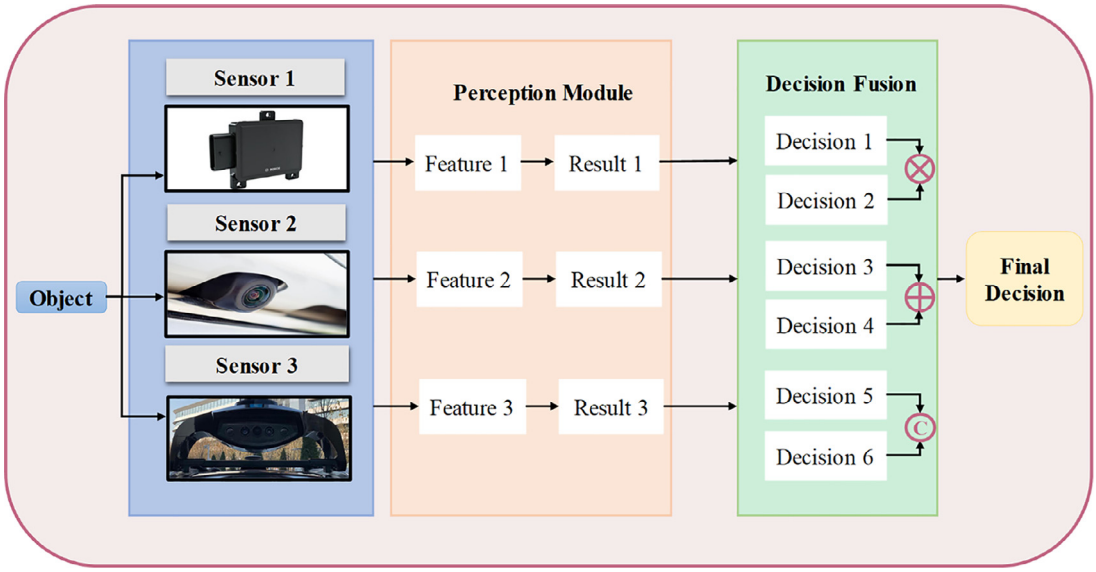


Fig. 5. Illustration of the decision fusion main structure.

5.1.3. Multi-stage Fusion

Early fusion methods utilize low-level data or feature information within the fusion algorithm framework, significantly reducing information loss during the fusion process. Theoretically, this can yield excellent results. Studies have shown that early-stage fusion produces better outcomes than mid or late-stage fusion (Huang and Hao, 2021). However, in the practical deployment of algorithms, early fusion is limited by its need for complex architectural design, higher computational power, issues with asynchronous timing and jitter-induced noise environments, and extended processing times. In contrast, late fusion requires only the detection results from upstream modules as input, thus relying less on the mid to early-stage modules and consuming less computational power. However, this approach results in underutilization of information from multiple sensors. In practice, the field currently favors a combination of multiple stages of fusion to complete perception tasks. The most common method combines feature-level and decision-level fusion. This approach typically utilizes data from one modality to generate candidate detection boxes, such as 2D candidate boxes from RGB data or 3D point cloud candidate boxes (Zhao et al., 2024). These candidate boxes are then merged with data of another attribute to produce the final object detection results. Additionally, there are numerous studies proposing the generation of candidate boxes in

BEV space (Liang et al., 2022; Vora et al., 2020; Wang et al., 2020; Wang et al., 2018; Zhou et al., 2023) to address the loss of spatial information during the transformation between 2D and 3D projections.

5.2. Fusion Methodology

With the advancement of multi-modal fusion, traditional approaches that merely concatenate the feature information of two or more modalities have become insufficient to meet the demands of highly automated AD systems. In recent years, this domain has seen a surge in the development of optimization algorithms and new algorithmic models. The following discussion categorizes and introduces common fusion methods for multi-sensor integration according to different technical approaches, demonstrating how these advanced technologies enable sophisticated functionalities in AD.

5.2.1. Data and Feature-based Approach

In the training process of models such as ML, a high-quality dataset is essential for enabling intelligent agents to better learn and understand the tasks they are designed to perform. Typically, preprocessing operations such as data cleaning, data transformation, and data augmentation are applied to the raw sensor data directly obtained from upstream sources. These operations convert the data into a standardized dataset suitable for model requirements. For perception tasks in AD, data-level preprocessing can effectively reduce the complexity of data processing by models, enhance model robustness, and improve generalization capabilities.

The initial step in data preprocessing is the cleansing of raw data, such as point clouds or RGB images. During actual operations, sensors mounted on AVs can be affected by factors such as vibrations and weather conditions, leading to issues like duplicate, missing, or anomalous data in the data collected from upstream sensors. A major challenge in data cleansing involves handling noise data, which can arise from complex environmental conditions such as extreme weather, high light intensity, and dense traffic flow. Wang et al. (2023) introduces a data preprocessing method based on Gaussian distribution that addresses spatial errors associated with 4D MMWRs. Moreover, during the model training process, a unique loss sampling mechanism can reduce the impact of noise on the learning process of intelligent agents (Dong et al., 2021). After processing, the standardized data is ready for training and learning in models. However, due to the significant computational demand required by advanced techniques such as deep learning, data is typically standardized or normalized prior to feature extraction, through methods such as Z-score standardization or Min-Max normalization. Feature encoding, such as One-Hot encoding, further allows ML algorithms to better understand and utilize the data.

Another crucial operation in data processing is data augmentation. Data augmentation refers to a series of random transformations applied to data to generate additional samples, thereby increasing the diversity of the dataset. These transformations and modifications can create a large number of new data points that are similar but not identical to the original data. This process can enhance the model's ability to generalize during training and learning. Moreover, overfitting—a phenomenon where a model performs better on the training set than on the test data—is commonly encountered in model training. Data augmentation can reduce model dependence on specific initial training samples by introducing new, diverse data, thereby mitigating overfitting.

Robustness of models is also a significant concern for researchers. Beyond operations such as flipping, scaling, and cropping, applying blurring or using generative models to add rain or fog to images can simulate real-world data diversity to some extent, thereby enhancing model robustness. For multi-sensor data in AD, given the limited number of objects in datasets, data augmentation is indispensable. However, when integrating heterogeneous sensor data, a major challenge is constructing fine-grained mappings between points or pixels. PointAugmenting (Wang et al., 2021) employs a cross-modal data augmentation algorithm that ensures pseudo-objects consistently follow both images and point clouds throughout the model's learning process. Typically, augmentation operations assume random selection of objects to ensure that the model learns accurate and generalizable information. This randomness significantly limits the consistency of data augmentation across different modalities. Current multi-modal augmentation methods involve slicing real objects' point clouds and image patches and applying them consistently across different scenarios to prevent misalignment. GraphBEV (Song et al., 2024) is a recently proposed fusion framework that uses graph matching to utilize nearby perceptual depth features and a module called GlobalAlign to correct misalignments between LiDAR and cameras. Additionally, DeepFusion (Li et al., 2022) proposes a multi-modal data augmentation alignment method that precisely spatially aligns RGB pixels from cameras with points from LiDAR through geometrically relevant operations such as rotation, effectively addressing the challenges of data augmentation in multi-sensor fusion.

After data cleaning and augmentation, data from different sensors are fed into their respective feature extraction models to obtain semantic and depth information in the form of feature maps or matrices. In recent years, with the maturation of deep learning techniques, the end-to-end approach to AD has been increasingly pursued by researchers. Traditional AD systems operate through multiple independent modules, such as perception and decision making, sequentially completing all tasks of AD. In contrast, end-to-end approaches map directly from input (upstream data from sensors) to output (decision and control commands of the AD agent), offering the advantage of simplifying the system architecture of AVs and reducing transmission errors, thereby enhancing overall performance. From the perspective of multi-sensor feature fusion, considering that the end-to-end model requires vast amounts of training data and computational power, many research efforts have foregone the addition of extra networks to transform voxels or pixels. Instead, they directly average each minimal granule as the feature representation for fusion. Simple geometric transformations and projections are used to fuse LiDAR and camera data at the feature level (Alfred Daniel et al., 2023; Huang et al., 2021; Kumar et al., 2020; Wang et al., 2020; Xiao et al., 2022).

This method of fusion effectively addresses the real-time challenges associated with end-to-end AD technologies. However, due to its overly simplified model architecture, this algorithm is only capable of fulfilling lower-level AD tasks such as distance detection and obstacle identification.

Contrarily, methods optimizing different spatial projections and aligning multi-sensor features, such as through viewpoint transformers, have been successively proposed (Das et al., 2024; Fadadu et al., 2022; Liu et al., 2023; Schramm et al., 2024; Vora et al., 2020; Wang et al., 2023; Zhao et al., 2024; Zhou et al., 2023). These transformers primarily convert point cloud features and image features into a unified feature space, typically the BEV space, to achieve alignment. The BEV space usually provides a top-down 3D perspective, which offers superior accuracy in perception tasks compared to the 2D RGB space. Furthermore, within the BEV space, the perception of object sizes and motion directions, as well as the richer 4D spatial information formed by integrating temporal data, enhance the algorithm's capability to address sensor viewpoint discrepancies and issues of point cloud sparsity and clutter encountered during fusion. Additionally, operations such as smoothing of feature maps (Yoo et al., 2020) have become an important approach to enhancing the effectiveness of feature fusion.

In addition to optimizing feature space fusion, some research has focused on designing affinity computation modules and data or feature association modules to achieve optimized fusion of different sensors. These algorithms leverage both decision and feature-level advantages, allowing the integration of information such as target location, speed, dimensions, and confidence obtained in the early stages of the model. Considering the real-time deployment needs of the algorithm, methods such as Dynamic Coordinate Alignment (DCA) have been proposed to minimize errors (Kuang et al., 2020). For tracking tasks, which demand high correlation, algorithms used for multi-object tracking like Kalman Filtering and DeepSORT are often employed in post-fusion multi-modal matching (Farag, 2021). Kalman Filtering is a highly versatile autoregressive filter that uses linear system state equations to optimally estimate the system's state based on its input-output data. DeepSORT, commonly used in traditional object detection and tracking tasks, utilizes detected object bounding boxes in each frame. It extracts features using deep learning network models and matches each object to those tracked in previous frames. During matching, factors such as feature similarity and motion consistency are considered. Thus, this method can accurately differentiate the similarity levels between different targets, providing a robust foundation for multi-sensor feature fusion. Additionally, methods like the Nearest Neighbor (NN) and Probabilistic Data Association (PDA) are also utilized for joint optimization (Huang and Hao, 2021). These algorithms are particularly effective in addressing perception errors caused by occlusions and the reduced accuracy of detecting distant targets in 3D space.

5.2.2. Optimization Approach of Fusion Network

With the widespread application of deep learning across various domains, the perception tasks of AD are also predominantly based on various network frameworks. Multi-sensor fusion, as a significant challenge and research direction in AD perception tasks, is increasingly attracting researchers to design specialized networks tailored to fit the requirements of the fusion process. These networks aim to effectively integrate and process data from different sensor modalities, enhancing the system's ability to perceive and interpret the surrounding environment accurately.

Xiao et al. (2022) combined RGB images and depth information as multi-modal data and designed a CNN framework based on conditional imitation learning that accommodates fused data. This demonstrated the advantages of multi-sensor fusion in perception tasks and laid a favorable research foundation for most Camera-Millimeter Wave Radar Fusion (CRF) efforts. Additionally, Xu et al. (2024) was the first to incorporate Driver Attention (DA) as a type of modal data into end-to-end AD. In the early stages of fusion, a DA prediction network was designed to obtain outcome information, providing new input features for subsequent fusion. The inclusion of the DA modality offers the model finer-grained perceptual information, enabling the AD agent to think like an experienced driver (Xu et al., 2024).

As a representative of cascaded fusion, Kuang et al. (2020) proposed a general fusion framework by designing an Deep Affinity Network (DAN) to enhance the performance of multi-modal data association. Similarly, Gu et al. (2021) developed a new cascading strategy for fusing LiDAR and camera data, addressing the issue of insufficient generalization capability in road detection models. Although cascaded neural networks can fuse features across different scales, the input features are transformed between 2D and 3D, which results in the loss of spatial information. Therefore, Wang et al. (2020) introduced a fusion network called Field of Fusion Network (FoFNet), which uses a pyramid network to extract and fuse features at multiple scales and then connects them across channel dimensions through deconvolution. Similarly, Zhou et al. (2023) begin by extracting features from radar point clouds and then achieves multiscale 2D image feature fusion through the image backbone model. For MMWR, Wang et al. (2023) proposed a center-based multiscale fusion method (CMSF), starting with object center regression followed by feature extraction at various resolutions to perform 4D MMWR feature extraction for subsequent fusion. For the same MMWR feature extraction, Lin et al. (2024) utilized an encoder-decoder network structure and designed a BEV feature extraction network called RadarBEVNet, consisting of a MMWR backbone network and a Radar Cross Section (RCS) BEV encoder. After extracting radar features, two dedicated modules are used to facilitate communication between the encoders, greatly improving efficiency and feature utilization. Additionally, given the finer granularity of multiscale information, Jiao et al. (2023) introduced a new fusion framework called MSMDFusion to facilitate information exchange between LiDAR and cameras using a Multi-Depth Unprojection (MDU) method to enhance feature quality at each angular level. This then integrates multi-modal features into a unified space to complete 3D object detection tasks with high utilization of multi-modal features. However, in real-world deployments, factors like adverse weather conditions can lead to sensor degradation and unreliable feature correspondence. Bijelic et al. (2020) proposed a fusion network that exchanges and adaptively fuses depth features from LiDAR and RGB cameras to address low robustness in adverse weather. For the robustness challenge of detecting distant targets, Zhang et al. (2021) proposed a fusion strategy called Faraway-Frustum. Relying on a 2D recognition network to detect objects and using point cloud data to locate objects in 3D space without affecting their shape proportions, this strategy achieves sensor fusion between LiDAR and cameras. Although these algorithmic models optimize multi-sensor fusion accuracy to a certain extent, lightweight models are essential in real-time AD perception tasks. One common strategy is partial network optimization, such as Dong et al. (2021), which designed a loss sampling mechanism that also mitigates the influence of noisy sensor measurements.

Additionally, for fusion methods in CNNs, Wang et al. (2018) designed a sparse non-uniform pooling layer for fusing multi-view features at any stage of the network. Another promising direction is using lighter network models (Wen and Jo, 2021) to replace complex multi-stream multi-branch models (Liang et al., 2022). Pang et al. (2020) introduced a low-complexity fusion framework called CLOCs, which utilizes matching geometric and semantic information trained to operate on the combined 2D and 3D output before NMS to efficiently fuse semantic and geometric detections. This method was further optimized for computational consumption and accuracy, leading to the Fast-CLOCs algorithm (Pang et al., 2022), which achieves nearly real-time high precision 3D object detection. In the real world, although the actual driving scenarios of AD are complex and varied, each dataset has its important RoI, so another way to reduce model complexity is to operate on the RoI. Existing literature also proves that this method achieves higher feature utilization during the fusion process (Xu et al., 2023; Yoo et al., 2020; Zhao et al., 2020; Zhou et al., 2023).

5.2.3. Attention-based Approach

In recent years, the attention mechanism has attracted increasing attention in artificial intelligence research. With the advent of the Transformer architecture, as introduced in the reference (Vaswani et al., 2023b). Replacing traditional neural network structures has led to a qualitative leap in the field of natural language processing. Subsequently, adaptations such as the ViT (Dosovitskiy et al., 2021b) have moved into the domain of computer vision, demonstrating exceptional performance in tasks such as image classification, object detection, and semantic segmentation. Initially, attention was applied at a basic feature level, using manually defined special weights to process specific parts of images (Xu et al., 2023; Yoo et al., 2020; Zhao et al., 2020; Zhou et al., 2023). The mentioned RoI in neural networks represent a specific form of attention. Detectors first identify the RoIs in the data, and then derive final outputs like classification categories or bounding boxes based on weighted processing of these RoIs. This method varies according to the downstream tasks and represents a top-down approach. However, this RoI-focused method is a very coarse form of attention. As research has progressed, an increasing number of researchers have adopted more sophisticated forms of attention such as channel attention, cross-attention, and self-attention. These methods achieve finer-grained attention, supporting a variety of tasks by more effectively discerning and emphasizing salient features within the data. This evolution marks a significant shift from basic attention implementations to more nuanced and dynamically adaptive mechanisms that enhance model performance across various AI applications.

As is well known in neural networks, the input and output of each layer correspond to multiple channels of features. Each different feature extraction channel corresponds to different information from the raw data, such as the category, shape, or color of an object in an RGB image. Depending on the task, the model's demand for the importance of different feature information varies. For example, lane detection tasks rely more on the color information of RGB images, while object detection depends more on shape and location. In such cases, channel attention assigns greater weight to the color channels for lane lines or the shape channels for vehicles and pedestrians. SENet (Hu et al., 2019) employs channel attention by performing global pooling on each channel, compressing it into a single value. This value for each channel becomes a vector dimension, which, after passing through a fully connected layer and transformed by a sigmoid function, becomes a value between 0 and 1. This value corresponds to the weight of each channel. Zhao et al. (2019) proposed a network architecture based on channel attention that uses the front camera and point cloud's visual frustum to generate 3D detection results. The SENet module then captures more important channel features to suppress those with less information. This method shows particularly good performance in 3D object detection tasks when point clouds are sparse. Additionally, considering the adaptive needs of end-to-end tasks, Zhang et al. (2022) proposed two fusion techniques based on channel attention mechanisms for different task scenarios, addressing the problem of exploring less feature information in auxiliary 3D object detection tasks. Channel attention can also be used as a module within a network to enhance performance. For lane segmentation tasks, Zhang et al. (2021) introduced a plug-and-play multi-modal segmentation network's channel attention module to obtain cross-channel interaction information, studying the contributions of different feature channels through varying weights.

Similar to channel attention, spatial attention and branch attention are also commonly used attention mechanisms by researchers. Spatial attention is entirely driven by the task and data at hand. This module involves weighting specific points or pixels on data or feature maps. On the other hand, branch attention arises because multi-modal fusion network structures often contain many branches. Simply concatenating these branches can sometimes lead to a decrease in accuracy. Branch attention combines features from different branches and assigns new, learnable parameter weights to them. Through this method, each branch in the network's training process is dynamically weighted, allowing for a more targeted and efficient learning and integration of features across the different modalities and functions within the network.

The rise of self-attention is credited to the outstanding potential and performance demonstrated by the Transformer, whose principal concept involves calculating the correlations among input data to facilitate the design of an encoder-decoder architecture (Vaswani et al., 2023b). As self-attention imposes no constraints on the input data type—applicable across temporal, spatial, or channel dimensions—many researchers have sought to adapt the Transformer, originally developed for the NLP domain, to suit their specific tasks (Wang et al., 2025).

The key to visual tasks is extracting correlations between pixels, and traditional CNNs use convolutional operations to extract feature maps with relatively limited receptive fields. Due to its ability to provide a global receptive field for input data, the performance of Transformer architectures often surpasses that of CNNs. Transformer-based vision models provide a foundational framework for vision-related tasks by dividing an image into multiple patches and arranging them into a sequence representation. In this classification task, an image is divided into multiple parts and arranged into a sequence representing the image. Each part's features become dimensions of the overall sequence feature vector. These features are then integrated and encoded by encoders using multiple self-attention mechanisms to extract various feature information, which fits well with the parallel computing capabilities of GPUs. The encoder-decoder network structure is thus widely applied to multi-sensor fusion (Bai et al., 2022; Das et al., 2024; Schramm et al.,

2024; Shao et al., 2022; Yang et al., 2022; Zhou et al., 2023). Unlike image data from cameras, point clouds are unstructured data. Each point in a point cloud can act as a Token subject to self-attention, focusing on the interrelations between points. However, due to the large number of points in point clouds, the computational load of self-attention can exponentially increase. It is necessary to perform hierarchical clustering of the point cloud (Qi et al., 2017), and then compute attention based on the clustering of each layer. For multi-sensor fusion tasks, the point cloud clustering algorithm also affects whether radar data can be effectively fused with other sensor data. Verucchi et al. (2020) proposed an optimized PointNet++ clustering method integrated into the Yolov3 (Redmon and Farhadi, 2018b) network architecture to achieve fusion of LiDAR and cameras, effectively solving the problem of real-time lag in AD systems deployed on embedded platforms. As one of the most commonly used attention structures in recent years, transformers are also widely used to link features from multiple modalities. D. Xu et al. (2024) proposed an LVAFusion module that uses prior Query (Q) to integrate feature representations of images and point clouds. This approach effectively captures the key features common to both sensors. Similar research efforts (Chitta et al., 2022; Fu et al., 2024; L. Liu et al., 2022; Prakash et al., 2021; Yang et al., 2022) integrate features from MMWR and cameras using self-attention, improving the interaction and fusion of multi-modal information. Notably, Wang et al. (2023) proposed an Interactive Multi-modal Fusion method (IMMF) that applies self-attention mechanisms to the intermediate layers of feature extraction, allowing the model to learn and exchange information about the features of each sensor's data. Moreover, Xu et al. (2023) uses self-attention within a single modality to enhance features of specific regions, then employs cross-attention to fuse information between modalities.

Different from self-attention, cross-attention allows each position in one sequence to attend to all positions in another sequence, focusing more on the interrelations between different types of data (Ke et al., 2021). Therefore, cross-attention is commonly used in multi-modal fusion tasks and is particularly suited for integrating data or feature information from different sensors into the same space in AD perception tasks. This mechanism effectively enhances the model's ability to synthesize and utilize diverse sensory inputs, improving decision-making and environmental understanding in complex driving scenarios.

To be specific, images from multiple viewpoints are typically first transformed into feature information through a network. Because the world coordinate system of BEV is particularly suitable for the perception tasks of AD systems, most BEV perception currently involves transforming image data, features, or detection results from cameras into BEV spatial coordinates for fusion. Therefore, the Q vectors in the attention mechanism originate from the BEV space. For each position in the BEV space, its features can be combined through weighted composition with vectors from image features using attention. For example, Lin et al. (2024) employed a cross-attention module to fuse multi-modal features from MMWR and cameras in the BEV space. Besides fusion in the BEV space, directly computing cross-attention between 3D point cloud space and RGB image features is also a commonly used method in this field (Li et al., 2022; Xie et al., 2020; Xu et al., 2023).

5.2.4. Other Fusion Approach

Cao et al. (2021) first investigated the safety issues of multi-sensor fusion in the AD domain. From the perspective of attacks on single or multiple fusion sources, a general multi-sensor fusion strategy was designed. Additionally, since the cost of LiDAR is significantly higher than that of MMWR and cameras, Zhao et al. (2024) proposed a Camera-Radar Knowledge Distillation (CRKD) strategy. Based on the BEV encoder-decoder structure, a teacher-student model was established. Knowledge distillation is a model compression method in ML. By designing a lightweight student network, it utilizes the information (knowledge) from a larger, more performant teacher model to train (distill) the student model, achieving more efficient and precise performance on the student model. CRKD implements the transition from LiDAR-Camera Fusion (LCF) to CRF by designing four specific distillation losses. This increases the model's efficiency while only losing a minimal amount of accuracy and saves substantial costs from a mass production deployment perspective by reducing sensor costs.

5.3. The Application of Multi-sensor Fusion in AD

According to the diverse demands of real driving environments, multimodal fusion methods have a certain degree of task orientation. Different fusion approaches or optimized fusion algorithms exhibit varying performance in complex and diverse situations. Researchers typically employ targeted optimization methods to enhance the performance of multimodal fusion in response to the various challenges encountered across different perception tasks. From a single-task perspective, the principal perception tasks faced by AVs include object detection, semantic segmentation, and tracking. Each of these tasks presents unique challenges that require specialized approaches to effectively integrate and analyze data from multiple sensor modalities. These optimization methods aim to improve accuracy, reduce latency, and ensure robust performance in diverse environmental conditions.

5.3.1. Single-task Model

Object Detection. AD agents require precise perception of the road environment to detect obstacle distributions, thereby ensuring safe and stable driving. Specifically, object detection involves accurately locating and predicting both static obstacles, such as traffic signs and buildings, and dynamic objects, such as pedestrians and other vehicles. Once an AV has predicted the object type, location, and size, it can utilize this information to perform subsequent decision making and control actions. Traditional object detection is categorized into 2D and 3D, corresponding to training data from camera images and LiDAR point clouds, respectively. After processing different data types, features are extracted, which then inform the design of detection heads to predict object position, size, category, and confidence, among other details. The model's performance is evaluated using the average precision (AP). Numerous object detection algorithms based on single sensors have been proposed in recent years. However, in real driving scenarios, the performance of perception models using single-sensor data can be compromised by obstacles, lighting conditions, and adverse weather.

To address these issues, Wang et al. (2023) leveraged the stable data distribution of MMWR under extreme weather conditions to fuse RGB images with MMWR point cloud data. Moreover, based on the camera's imaging principle, information in front of the vehicle converges towards a vanishing point on the road. Objects near this point have very small bounding boxes in the image, and the corresponding distant point cloud data is sparse. To resolve this challenge, Zhao et al. (2020) generated object proposals using LiDAR data and performed complementary fusion in RGB space, ingeniously extracting multi-scale feature information in the RoI using CNNs. Additionally, works like Jiao et al. (2023), Wang et al. (2021) and Zhou et al. (2023) optimized the feature extraction fusion module, allowing multimodal features to be integrated into a unified feature space with different granularities to achieve 3D object detection tasks.

Semantic Segmentation. Semantic segmentation has emerged as a significant research area in computer vision recently, demanding high-accuracy inference from agents to perform dense predictions and infer labels for each pixel, aiming to effectively segment fine boundaries of different objects. In the realm of AD, semantic segmentation primarily distinguishes between drivable and non-drivable areas. Beyond typical obstacles like pedestrians, vehicles, and buildings that need to be avoided, a variety of objects containing semantic information useful for driving—such as speed bumps, no-parking zones, and guide lines—emerge in complex traffic environments like city streets, highways, steep slopes, and tunnels. The task of semantic segmentation is to transform these diverse images into color blocks carrying semantic information, then identify the semantic category of each block and label every pixel. The mean IoU is commonly used as an evaluation metric to assess model performance in segmentation tasks. Although RGB images, rich in semantic information, have yielded considerable success in 2D segmentation (Rubio et al., 2013), 3D segmentation based on point clouds does not match the accuracy or processing speed of camera-based segmentation. In response, researchers have recently turned to multimodal fusion methods to effectively integrate the rich semantic features of 2D space with the positional and categorical features of 3D point clouds, thus improving segmentation accuracy. Zhao et al. (2024) developed a cost-effective method to align features from LiDAR data and camera data, enabling the model network to utilize image context information from coarse to fine detail for semantic segmentation. Similarly, Schramm et al. (2024) uplifted visual features to BEV space before performing CRF. Additionally, for specialized semantic segmentation tasks like road detection, Gu et al. (2021) employed a cascaded approach for fusing LiDAR and camera data. Specifically, this method first uses a LiDAR segmentation module to segment road points in the view, upsampling the sparse point cloud feature map into a dense road prediction. This result is then cascaded with RGB images to achieve accurate road estimation. Furthermore, for lane line segmentation, because of the particularity of lane lines, approaches like Zhang et al. (2021) use channel attention mechanisms to enhance fusion performance, which is one of the mainstream methods in the field of multi-sensor fusion in segmentation.

Tracking. Tracking tasks in the domain of AD play a crucial role in vehicle obstacle avoidance and subsequent decision-making processes. The primary operation involves using the initial frame's target position and other information within a time series to predict the target's location and size in subsequent frames. Traditional tracking algorithms generate multiple candidate boxes for the next frame based on the initial target's bounding box, extract features from these candidates, and rank them using a scoring mechanism. Subsequent updates to the observation model adapt to changes in the target's state over the series, selecting the optimal decision result as the prediction output. In recent years, with the explosive improvements in accuracy brought by deep learning across various tasks, coupled with the advent of various optimized filtering techniques, multi-object tracking has become one of the most challenging research problems in the field. Due to the real-time requirements of this perception task and its focus on studying the correlations between motion states, multi-sensor fusion, as proposed in works like Huang and Hao (2021) featuring affinity modules and data association modules in an end-to-end network model, demonstrates exceptional performance in applications.

5.3.2. Multi-task Model

With the rise of end-to-end research in AD, a significant portion of researchers have focused on reducing model complexity. Traditional AD systems consist of multiple task modules, each corresponding to one or several algorithmic networks or modules. This configuration significantly increases memory usage and reduces operational efficiency during deployment and application, thereby diminishing the real-time perception performance of AD agents. While parallel computation can mitigate time costs to some extent by leveraging GPUs, it increases vehicle production costs from a mass production perspective. For a complete AD perception system, tasks such as object detection, semantic segmentation, depth estimation, and lane detection are all essential in practical applications. Therefore, designing multi-task collaborative processing solutions plays a crucial role in the development of ADAS (Cakir et al., 2022; Elhassan et al., 2024; Kumar and Mondal, 2021; Wang et al., 2024; Zakaria et al., 2023). Additionally, real-time information processing efficiency is one of the core elements in AD systems. Due to the complexity and variability of driving scenarios, the perception system must complete environmental sensing, data processing, and result output within a very short time. Facing the complexity and diversity of information processing, meeting real-time demands has become a key challenge in the design of AD systems. Moreover, the dynamic nature of the environment presents a significant challenge to the performance of multi-task networks. Specifically, the variability of the environment requires multi-task networks to have the ability to quickly adapt to different scenarios. In this context, the network must be capable of handling complex environmental information while maintaining high real-time performance. In response to this challenge, multitask models have emerged as an effective solution.

The Multi-task Model (MTM) is a deep learning approach derived from the Multi-task Learning (MTL) algorithm, which enhances the model's generalization ability and overall performance by learning multiple related tasks simultaneously. MTM abandon the approach of using multiple models for different tasks and instead establish relationships among the requirements of various tasks to construct a unified structure that serves diverse needs. Unlike traditional single-task learning, which optimizes the model for a single

task, the MTM improves the performance of multiple related tasks by sharing features between them. During the network learning process for different tasks, each task can focus on its own specific details in separate branches, while sharing common features from the shared network. In MTM, different task flows have shared parameters with varying weights, allowing the model to be constrained by tasks with a regularization effect during training. This approach significantly reduces the risk of overfitting, especially when training samples are limited. The tasks can leverage the available data to improve generalization and address the challenges of each task in real-world applications. From the perspective of computational resources and efficiency, the shared components effectively reduce the number of parameters in the model, lowering its complexity while maintaining accuracy. Moreover, compared to an AD perception system with multiple single-task modules, MTM does not require independent network structures for each task. This results in reduced memory usage, increased computational efficiency, and the elimination of redundant calculations. For instance, lane detection typically relies on edge detection and geometric model recognition and segmentation, while tasks like pedestrian detection and traffic sign detection focus more on shape, color, and other features. These similar visual characteristics can be learned and enhanced through shared feature extraction layers in the network structure, improving performance for each task while maintaining task-specific accuracy (Yang et al., 2024). Tasks related to object detection can effectively learn the spatial locations and shape features of different objects. Based on this, transfer learning can share this knowledge and implement the MTM design between different task types by setting reasonable loss functions. In different scenarios, the model can also dynamically adjust the weights of different tasks using multiple shared network layers to fit task requirements, further enhancing the flexibility and scalability of ADAS.

Although MTL has demonstrated good performance in many fields, it also faces some challenges, particularly in the similarity between tasks and the design of shared information. If the correlation between multiple tasks in the model is insufficient or if there is a significant conflict in the shared features between tasks, it may lead to poor learning of shared parameters, and even cause negative transfer (NT) issues (Luo et al., 2024). NT occurs when there is a significant difference in gradient directions between different task flows during the learning process. This can cause the model's parameters to oscillate during training or result in one task's gradient excessively dominating the optimization process. Additionally, simpler tasks may converge quickly and reach a smaller loss value, causing the shared representation to prematurely fit these tasks, which reduces the model's ability to learn more complex tasks and may lead to overfitting. Therefore, how to balance the weights between different tasks and avoid having a dominant task that drives the entire training process is a problem that many researchers are working to solve (Chen et al., 2018; Cipolla et al., 2018). Liu et al. (2023) introduced a multitask, multimodal fusion framework called BEVFusion. This model, serving as a generic BEV representation space fusion framework, has been proven to be task agnostic. Its core idea is to transform feature information from different sensors into the BEV space and optimize BEV pooling, ensuring that the features required for different tasks are effectively utilized within the fused feature space.

The above methods show strong results on standard benchmarks. In practice, however, attention-heavy fusion and multi-task perception are shaped by latency, memory, training stability, and supervision cost.

Attention mechanisms are now a key part of the design of many new autonomous driving perception models because they make it possible to represent global context and align different types of data in a flexible way. Vision Transformers and hierarchical variants enhance scene understanding by enabling the capture of long-range dependencies that are challenging for local convolutions to process. In multi-sensor fusion, cross-attention proves particularly effective for associating camera features with lidar representations within a shared bird's-eye view (BEV) space. These advantages are most apparent in complex urban settings, where safety-critical cues may be spatially distant yet behaviorally crucial. Concurrently, deployment constraints limit the practical applicability of attention-heavy architectures. Standard self-attention suffers from poor scalability with the increasing token counts, amplifying memory pressure and latency under high-resolution multi-view inputs. Furthermore, when the target domain diverges from the pre-training distribution and data is scarce, Transformer models typically demand greater data volumes and meticulous optimization to mitigate training instability.

Multi-task learning is widely used in the operational perception stacks to reduce redundancy and take advantage of shared structure among tasks such as detection, lane segmentation, and drivable area estimation (Wang et al., 2025). The joint models can improve computational efficiency and representation sharing. Actual systems such as YOLOP show that a single backbone network within multiple heads can achieve real-time performance while maintaining competitive accuracy. However, multi-task learning introduces optimization conflicts that are directly relevant to safety critical performance. This problem motivates explicit balancing strategies, including gradient normalization and uncertainty-based loss weighting, and gradient surgery to reduce conflict in the process of back propagation. Multitasking learning also increases the need for supervision, because large data sets with consistent labels across tasks are still limited.

Beyond onboard perception, recent systems further extend awareness through cooperative agents and infrastructure via V2X communication.

5.3.3. Cooperative Perception with V2X Communication

Although conventional multi-sensor fusion mainly focuses on integrating heterogeneous information from onboard sensors such as cameras, LiDAR, and MMWR, recent studies have further extended this paradigm through V2X-enabled cooperative perception (Gu et al., 2022). In this context, V2X should not be regarded as an alternative to onboard multi-sensor fusion, but rather as a complementary extension that incorporates external sensing information from nearby vehicles and roadside infrastructure. By combining onboard perception with shared environmental information, cooperative perception can further improve situational awareness and perception robustness in complex traffic environments.

Vehicle-to-everything (V2X) communication connects the ego vehicle with nearby vehicles, roadside infrastructure, vulnerable road users, and network services. When V2X messages are integrated into the perception system, the sensing range is no longer

bounded by the ego vehicle's line of sight. Cooperative perception can recognize occluded targets in advance, such as pedestrians hidden behind trucks, or pedestrians detected by upstream vehicles or intersections equipped with sensors.

Recent surveys describe a common workflow that includes perception generation at each agent, information selection or compression, communication, alignment, and fusion at the receiver. This framework illustrates how V2X supports downstream planning and control by reducing blind spots and lowering the risk of late hazard recognition (Cui et al., 2022). Practical deployment introduces constraints that directly shape the algorithm design. First, benefits depend on penetration rate and infrastructure availability (Wang et al., 2024). Limited adoption reduces coverage and makes performance uneven across regions. Second, communication quality matters. Delay, packet loss, and bandwidth limits can degrade cooperative perception if shared information arrives late or incomplete. Recent research emphasizes delay-sensitive designs, efficient message representation and edge assisted architecture to reduce end-to-end delay. The third challenge is alignment. Shared information must be transformed into the ego frame and synchronized in time. Localization errors and asynchronous transmissions can cause spatial or temporal mismatches and fusion errors in fast traffic (Cheng et al., 2025; Wang et al., 2026). Finally, security and trust issues should not be ignored, because collaborative awareness relies on authenticated and reliable information, which is particularly important in security critical scenarios. These considerations have prompted the development of standardized message formats, robust synchronization mechanisms and verification methods, as well as the development of data sets and benchmark tools that reflect real multi-agent conditions.

The cooperative perception with V2X extends situational awareness beyond the physical limits of on-board sensors. Cameras and LiDAR can be blocked by buildings, large vehicles, or road geometry, especially at intersections and in dense urban traffic. V2X allows nearby vehicles and roadside units to share detected objects or perception features, so the ego vehicle can obtain non-line-of-sight cues that could not be obtained. Safety benefits come from earlier and more reliable hazard recognition. When an occluded pedestrian, bicycle, or suddenly brake vehicle is detected by another agent, the ego vehicle can react before the hazard enters its own sensor field of view. In this sense, V2X complements multi-sensor fusion by adding external viewpoints and redundant observations. It can increase detection confidence, reduce missed detections in blind spots, and support safer planning through more conservative and informed decisions in complex traffic interactions (Wang et al., 2023).

6. Discussion

6.1. Limitations of Cameras

Because AVs must continuously perceive semantic information from their surroundings, especially in complex road environments, cameras are essential sensors for high-level autonomous AD tasks. As the "eyes" of AVs, cameras capture RGB image data that contains rich information about the road environment. With advancements in deep learning and computer vision, AD systems installed in AVs can utilize the information from upstream RGB images for downstream perception tasks and even for the overall decision-making and control of the vehicle. This is particularly critical for understanding obstacle categories, traffic light signals, and traffic-sign semantics during driving. In other words, cameras are fundamental for the interaction between AVs and the real-world driving environment as they essential visual cues for these perception sub-tasks. Additionally, compared to other sensor types, cameras can significantly reduce production costs for AVs, making them the preferred sensor for many vehicles equipped with lower-level AD. Although cameras are widely used in the field of AD as the dominant sensing modality, it still has the following shortcomings:

Insufficient Model Interpretability

Deep learning-based methods, despite their effectiveness, often operate as "black boxes," with limited interpretability. This lack of transparency can hinder trust, verification and safety validation of autonomous driving systems, making it imperative to develop techniques that offer greater insight into model decision-making processes.

High Computational Requirements

Many learning-based methods deploy complex neural networks that demand substantial computational power. This requirement makes these methods less feasible for integration within the current generation of ADAS, necessitating the development of more efficient, lightweight ML technologies.

Inadequate Detection of Small Objects

The detection of small objects remains a significant challenge in camera-based autonomous systems. Enhancing detection accuracy may require the integration of multimodal fusion techniques that leverage data from various sensor types to improve the overall detection performance.

Optical Distortion

The deviations in lens geometry, particularly with ultra-wide lenses like those used in fisheye cameras, can lead to optical distortions such as barrel, pincushion, and mustache distortions. These distortions can inaccurately represent the environment, misplacing detected obstacles or features, thus necessitating intrinsic calibration of cameras to correct these distortions and accurately estimate camera parameters.

Susceptibility to Environmental Conditions

The quality of images captured by cameras can be severely affected by adverse weather conditions, such as snow, rainstorms, or intense sun glare. This sensitivity requires additional computational resources to process and analyze image data accurately under varying environmental conditions, presenting a substantial challenge for the deployment of camera-based systems in all-weather scenarios.

Although camera-based perception systems provide rich semantic information and cost-effective sensing capability for autonomous driving, they still exhibit inherent limitations in environmental adaptability, depth perception, small-object detection, and computational efficiency. Camera sensors are highly susceptible to adverse weather, illumination variation, and optical distortion, which may significantly affect perception reliability under complex driving conditions. These limitations indicate that relying solely on camera-based perception remains insufficient for achieving robust and reliable autonomous driving perception, thereby further motivating the development of multi-sensor fusion approaches that integrate complementary sensing information from heterogeneous sensors.

6.2. Limitations of Radars

MMWR is one of the radars commonly used in AD systems. Its advantage lies in its strong adaptability to complex environments, characterized by its capability to operate under a wide range of weather and lighting conditions. MMWR can function less affected by many adverse weather conditions, maintaining relatively stable performance even under extreme weather such as rain, snow, or sandstorms. Additionally, due to MMWR's strong penetration ability and excellent performance in detecting range and radial-velocity measurement in terms of distance and speed, it is often used for long-range object detection. From an industrial perspective, compared to other types of radars used for high-precision perception tasks in AD, such as LiDAR, MMWR has become a widely adopted sensor in mass-produced AVs due to its lower cost.

Despite its advantages, MMWR still has several major limitations:

Environmental Impact

Despite its superior ability to penetrate fog and smoke, MMWR may experience performance degradation in heavy rain, snow, or extreme weather conditions (Zang et al., 2019). Environments with high humidity levels can lead to signal attenuation, thereby affecting detection accuracy and range.

Material Limitations

MMWR exhibits limitations in detecting certain materials, such as nonmetallic objects or transparent substances like glass, which may not be effectively identified.

Angle and Direction Constraints

The detection capability of MMWR diminishes for objects moving laterally or diagonally due to the beam's strong directionality. This results in inferior detection performance for non-radial or laterally moving targets compared to linearly approaching ones.

Resolution and Accuracy

MMWR has lower spatial resolution than LiDAR, restricting its utility in detailed environment sensing. Although technological advancements can somewhat mitigate this limitation, MMWR still remains less capable of capturing fine-grained object geometry.

Multi-path Interference

In complex urban settings, MMWR beams may reflect off multiple surfaces, causing multi-path interference that undermines radar signal accuracy.

Size and Cost

Despite advancements that have reduced size and improved cost-effectiveness, high-performance MMWR systems remain relatively expensive and bulky, constraining their widespread adoption.

Electromagnetic Interference

Operation at high frequencies exposes radar systems to electromagnetic interference from other devices within the same frequency band.

In contrast, for tasks requiring higher precision in object detection, especially dynamic object detection and other high-level AD tasks, LiDAR is widely used due to its excellent performance. Compared to MMWR, LiDAR offers significantly higher measurement accuracy. This performance advantage is particularly important in perception tasks. LiDAR is well-suited to assist ADAS in completing very precise and safety-critical driving tasks.

Despite these advantages, LiDAR still has the following limitations:

Line-of-sight Dependency

LiDAR functionality hinges on objects within its line of sight, rendering it vulnerable to obstructed views. Adverse weather conditions, such as dense fog, heavy rain, or snow, can compromise LiDAR performance.

Cost Constraints

LiDAR technology is expensive, constraining its application in large-scale scenarios. Despite advancements in reducing costs, expense remains a limiting factor.

Resolution and Precision

LiDAR exhibits limited resolution and accuracy in capturing distant and small targets, potentially impeding target recognition and tracking.

Optical Sensitivity

LiDAR's efficacy is influenced by the optical properties of targets, including reflectivity and surface materials, impacting its detection and measurement capabilities.

Data Processing Complexity

LiDAR generates substantial data volumes, necessitating sophisticated data processing algorithms to extract pertinent information and conduct a thorough analysis.

Multi-path Interference

LiDAR beams may reflect off multiple surfaces in intricate urban settings, resulting in multi-path interference that compromises measurement accuracy.

In summary, although MMWR and LiDAR sensors provide significant advantages in environmental robustness, geometric perception, and depth-aware sensing compared with camera-based systems, each sensing modality still exhibits inherent limitations in perception capability, sensing resolution, environmental adaptability, and deployment cost. MMWR performs well under adverse weather conditions but suffers from limited spatial resolution and semantic representation capability, while LiDAR provides accurate geometric information yet remains sensitive to environmental conditions and computational complexity. These complementary characteristics further demonstrate that relying solely on a single sensing modality is insufficient for achieving robust and reliable autonomous driving perception under diverse real-world scenarios.

6.3. Challenges and Future Trends

In general, multimodal perception technology, especially multi-sensor fusion based cooperative perception in the domain of AD, is experiencing significant growth and holds immense potential for future research and application.

6.3.1. Diverse Sensing Systems

In addition to the three main sensors discussed in this paper, ultrasonic radar, Global Positioning System (GPS), and other sensors are also used for specific AD tasks. Considering the emergence of end-to-end AD systems, a robust sensor system needs to support not only AD perception tasks but also control, planning, and other tasks. Sensors such as the Inertial Measurement Unit (IMU) need to be integrated into the sensor system. As the variety and performance of sensors increase, the diverse sensor combinations offer more options for the construction and application of AVs. Thus, designing a reasonable sensor system has become a significant challenge in AD research. Different sensors exhibit varied performance in terms of cost, data modality, spatial resolution, and so on. Future research needs to explore the optimal combinations of diverse sensor systems, aiming to balance cost reduction with performance and interference resistance.

6.3.2. Calibration of Multiple Sensors

The calibration of multiple sensors stands as a pivotal procedure in ensuring the consistency of data from various sensors, which involves the measurement and estimation of both external and internal parameters, as well as processes such as data alignment and error correction. Proper calibration can enhance the performance and reliability of multi-sensor systems, thereby achieving more precise and reliable environmental perception. Despite the complexity of the calibration process, the obtained calibration results often lack the required level of accuracy. Consequently, there exists a necessity for additional investigation aimed at enhancing the simplicity and precision of calibration for both radar and camera systems.

6.3.3. Robustness in Adverse Environments

The robustness of multi-sensor fusion in extreme environments pertains to its resilience and reliability under adverse conditions like inclement weather, low light, and high-speed motion. The robustness is critical for pivotal applications such as AD, where vehicles must safely perceive and comprehend their surroundings across diverse environmental scenarios. Further exploration is required to delve into the stability research of fusion technology in AD under extreme environments.

6.3.4. Computation Efficiency

The real-time performance of radar and vision fusion within the AD context delineates the swiftness and adaptability of this technology in assimilating and interpreting perceptual data. Such expeditiousness assumes critical importance for AD systems, enabling timely perception and understanding of the surrounding milieu, thereby facilitating instantaneous decision-making and actions to ensure safe vehicular navigation. In AD scenarios, radar and vision sensors typically concurrently accumulate copious datasets, including images and point clouds. Thus, ensuring the expeditious processing and comprehensive analysis of this acquired data remains imperative.

6.3.5. Cooperative Platoon Control

In the realm of autonomous fleet coordination, the integration of radar and vision technologies facilitates streamlined collaboration and intelligent scheduling among vehicles within the platoon. The integration aims to bolster operational efficiency and safety measures. Leveraging real-time perception and data analytics, fleet managers can enhance monitoring capabilities concerning vehicle positions, driving velocities, and traffic dynamics, which enables prompt adjustments to fleet routes and driving strategies, ultimately optimizing fleet operations.

6.3.6. Smart Infrastructure Services in Transportation 5.0

In the forthcoming advancement of smart infrastructure services in transportation 5.0, the amalgamation of radar and vision technologies holds promise for diverse applications including urban traffic management, intelligent traffic signal systems, and smart parking solutions, which promises to optimize urban traffic flow and road-network capacity, mitigate congestion, and reduce accident rates. The aforementioned fusion methodologies can enhance smart parking systems, expediting parking-space detection and localization for vehicles and enhancing overall parking efficiency. Furthermore, the large-scale deployment of multimodal perception technology extends to urban security surveillance, environmental monitoring, and urban planning, offering a solid foundation for the development and governance of smart cities.

7. Conclusion

To comprehend recent advancements and stimulate new research inspiration on this crucial topic, as well as to summarize and interpret the strategies of enhancing perception performance in the context of AD, this survey systematically reviews the perception algorithms based on different types of sensors, with particular emphasis on state-of-the-art multi-sensor fusion approaches, and outlines future directions, thereby providing a comprehensive overview of current research progress.

Firstly, this survey analyzes commonly used open-source datasets in this field, summarizing and organizing them based on the types of sensors used, supported perception tasks, and data diversity. By systematically reviewing representative datasets, we conclude that due to the diversity of real-world driving scenarios, single-modal data cannot effectively facilitate the development of AD perception systems. Therefore, multi-sensor fusion has emerged as a highly promising research direction in AD perception. Secondly, it reviews existing sensor-based perception algorithms from two aspects. One aspect focuses on camera-based perception systems. This paper discusses different camera-based perception methods from traditional approaches to ML and deep learning, pointing out the challenges and issues faced by cameras in terms of depth information and detection accuracy. The other aspect reviews cutting-edge research based on MMWR and LiDAR. Introducing the working principles and application scenarios of both radars, it highlights the limitations of radar-based methods in perception tasks and potential future research directions. Given the limitations of single sensor-based methods, the review mainly analyzes and summarizes advanced multi-sensor fusion technologies. It introduces the common fusion stages in this research field, leading to several classic fusion methods in both the pre-fusion and post-fusion stages. Particularly, the article proposes a classification method based on different fusion methods. we also summarize and categorize the recent fusion investigations from the perspective of fine-grained data enhancement approaches to proposing holistic fusion frameworks. Additionally, by linking the refined perception sub-tasks targeted by diverse fusion technologies, this review explores the intrinsic mechanisms of distinct strategies. Thereby, we identify the advantages and disadvantages of different methodologies for solving specific issues and challenges, and suggesting future research directions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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